Lecture 08

Multiple Market Failures and Second-Best Policies

Ivan Rudik
AEM 4510

Roadmap

1. What happens when we have another market failure like market power?
2. How do second-best policies like output taxes or intensity standards work?

Market power and pollution

Market power

Lets consider two extreme cases to understand whether and how market power matters

1. Perfect competition
2. Monopoly

Market power

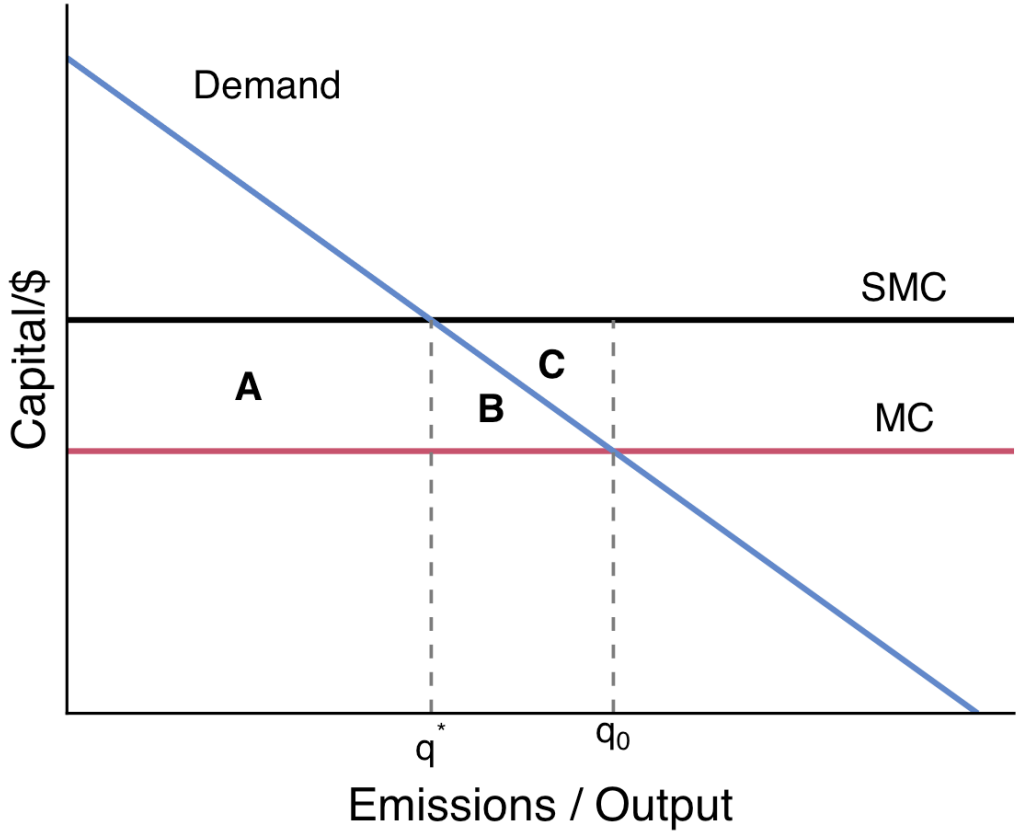
Lets consider two extreme cases to understand whether and how market power matters

1. Perfect competition
2. Monopoly

In both cases we will assume that:

1. Marginal costs of production are constant MC
2. Emissions are perfectly proportional to output, so each additional unit of output creates the same marginal damage and gives us constant social marginal cost $SMC = MC + MD$

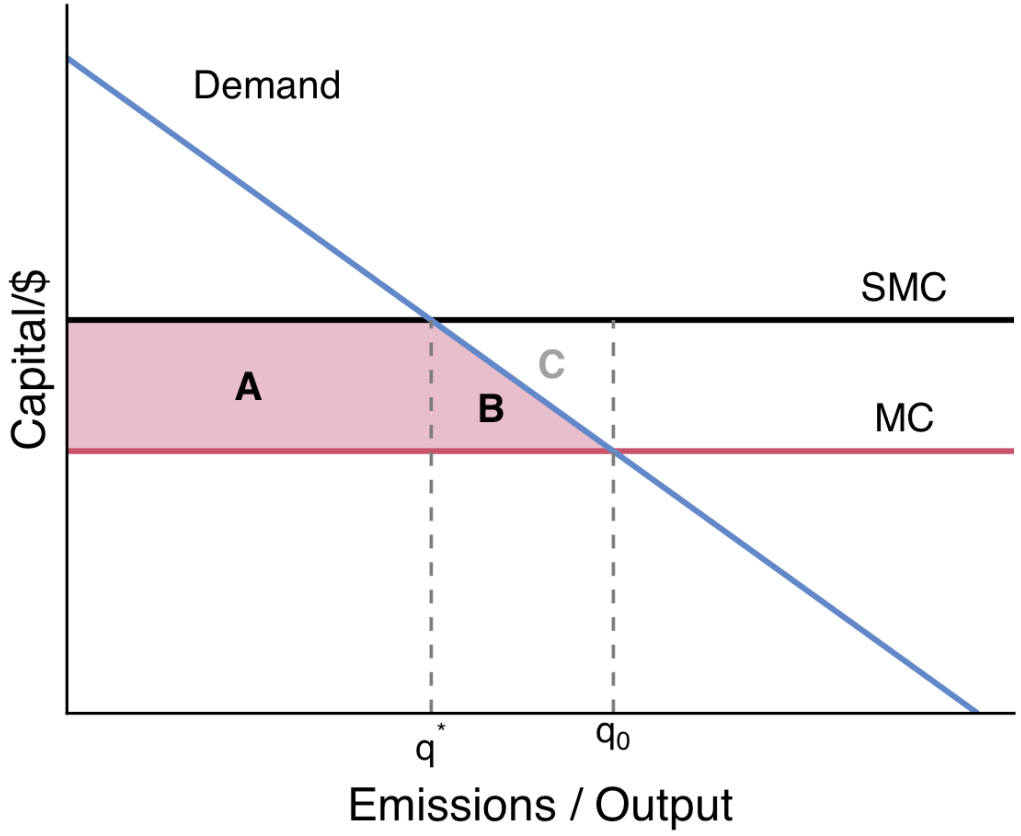
Perfect competition



A tax of $SMC - MC$ moves output from q_0 to q^*

Track the welfare effects using areas A, B, and C

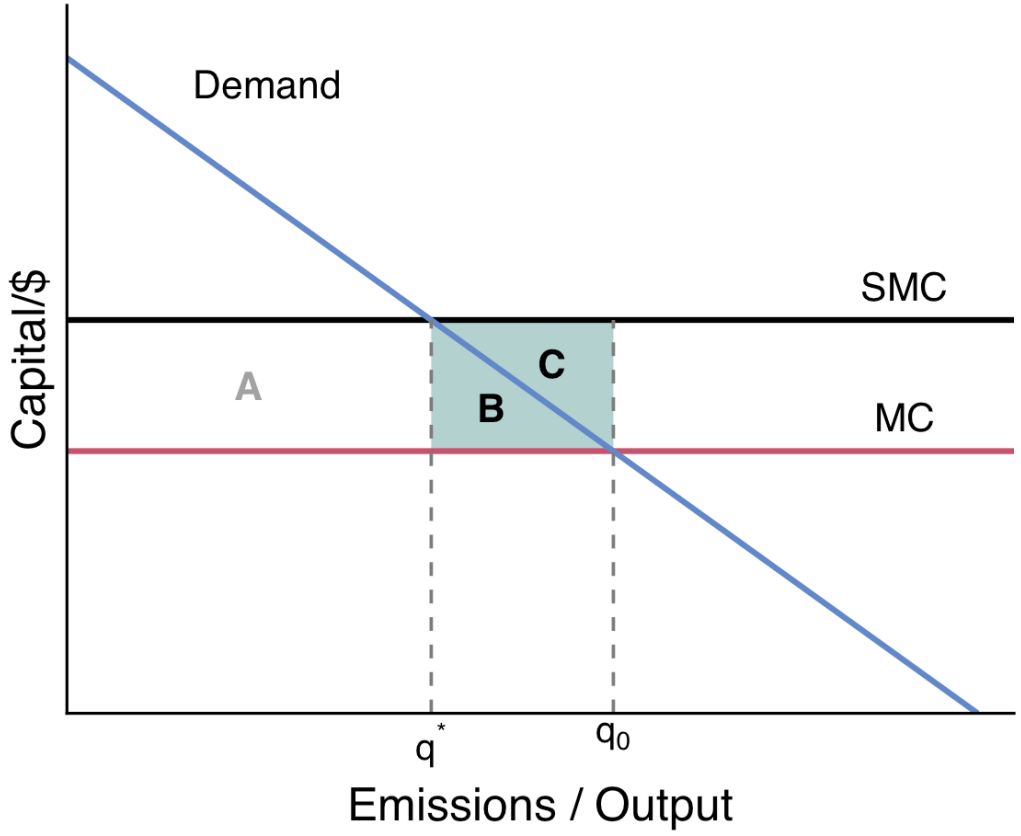
Perfect competition



A tax of $SMC - MC$ moves output from q_0 to q^*

Loss in CS: $-(A + B)$

Perfect competition

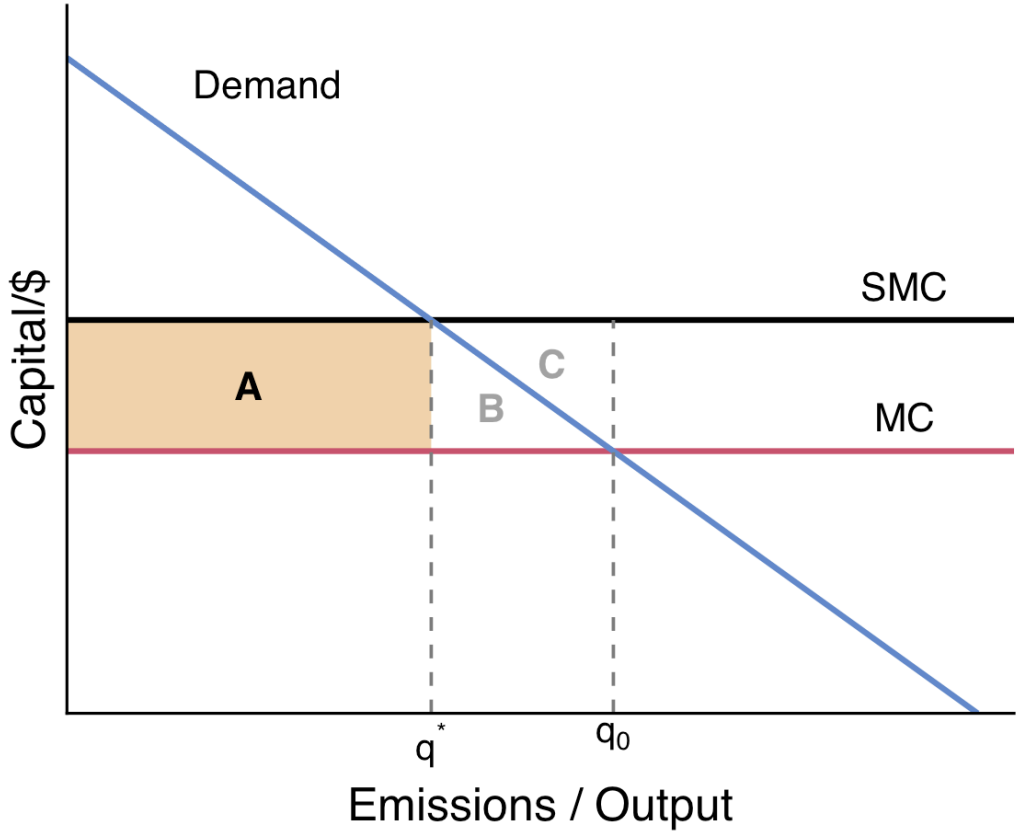


A tax of $SMC - MC$ moves output from q_0 to q^*

Loss in CS: $-(A + B)$

Avoided damages: $B + C$

Perfect competition



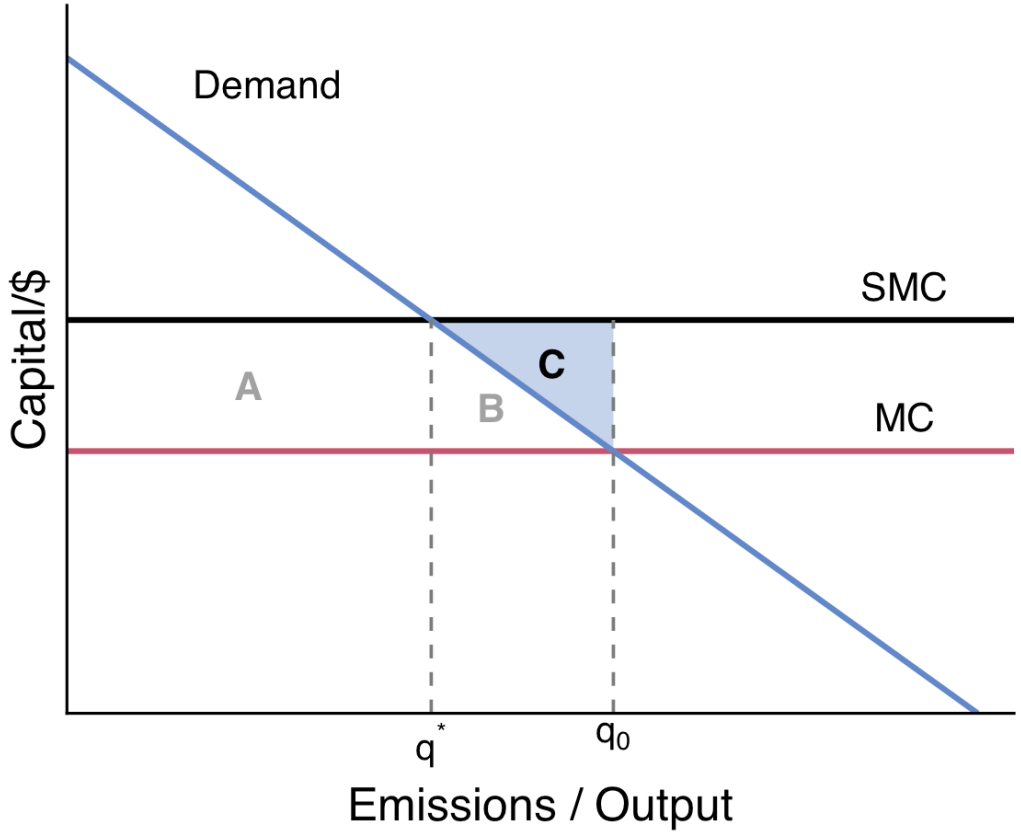
A tax of $SMC - MC$ moves output from q_0 to q^*

Loss in CS: $-(A + B)$

Avoided damages: $B + C$

Tax revenue: A

Perfect competition



A tax of $SMC - MC$ moves output from q_0 to q^*

Loss in CS: $-(A + B)$

Avoided damages: $B + C$

Tax revenue: A

Net gain:

$$-(A + B) + (B + C) + A = C$$

Monopoly

Now consider a monopolist with the same marginal cost and marginal damage structure

Monopoly

Now consider a monopolist with the same marginal cost and marginal damage structure

What is the difference with a monopolist?

Monopoly

Now consider a monopolist with the same marginal cost and marginal damage structure

What is the difference with a monopolist?

The monopolist can set the price

Monopoly

Now consider a monopolist with the same marginal cost and marginal damage structure

What is the difference with a monopolist?

The monopolist can set the price

This means that the MR curve lies beneath the demand curve

Monopoly

Now consider a monopolist with the same marginal cost and marginal damage structure

What is the difference with a monopolist?

The monopolist can set the price

This means that the MR curve lies beneath the demand curve

Why?

Monopoly

Now consider a monopolist with the same marginal cost and marginal damage structure

What is the difference with a monopolist?

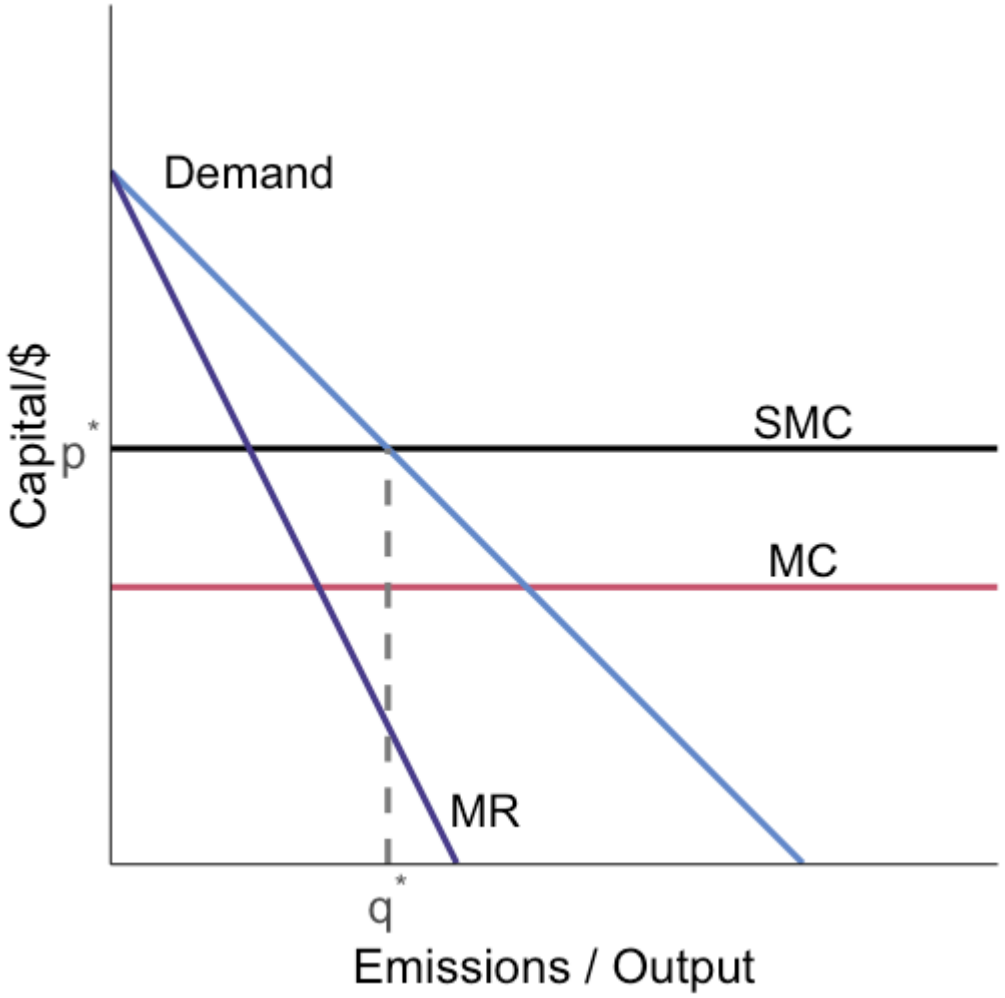
The monopolist can set the price

This means that the MR curve lies beneath the demand curve

Why?

The monopolist accounts for how additional output lowers the market price on inframarginal units

Monopoly

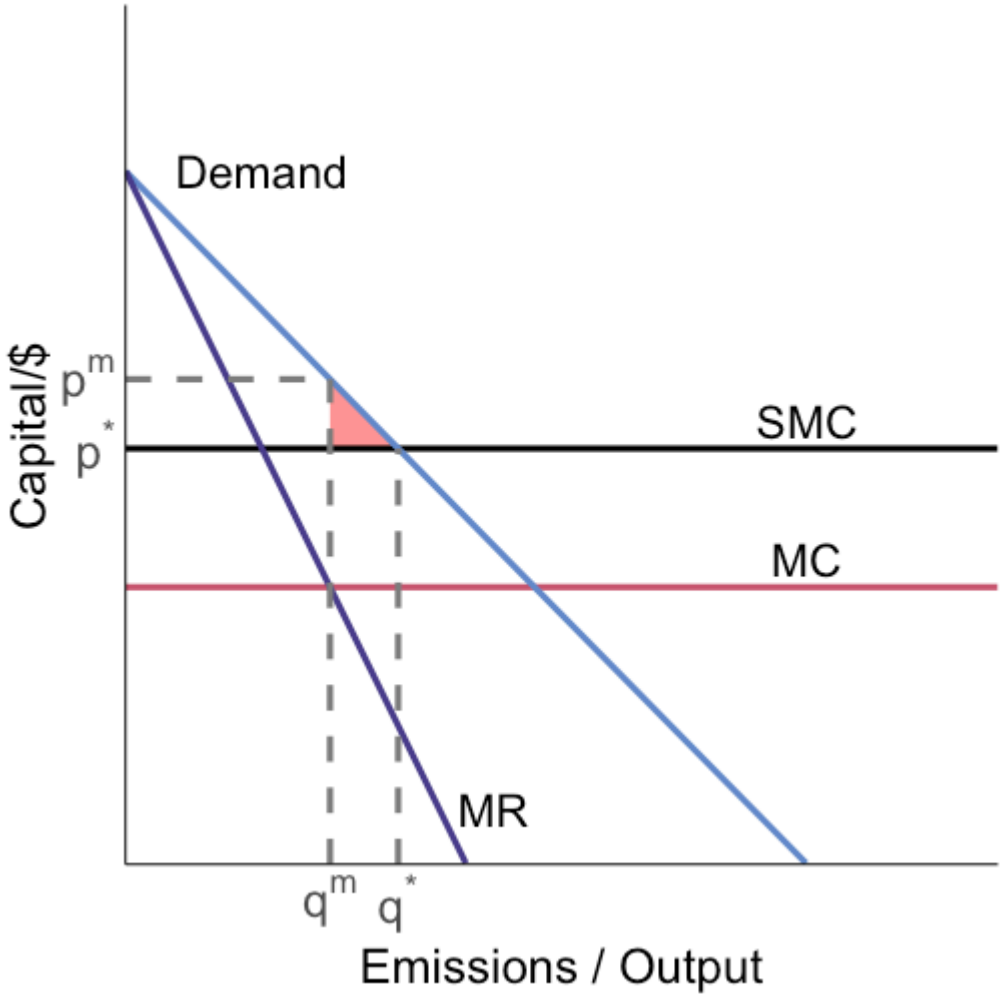


The socially efficient allocation is where social marginal cost is equal to the social marginal benefit

This is where SMC crosses the demand curve: (q^*, p^*)

What is the welfare outcome under the unregulated monopolist outcome?

Monopoly

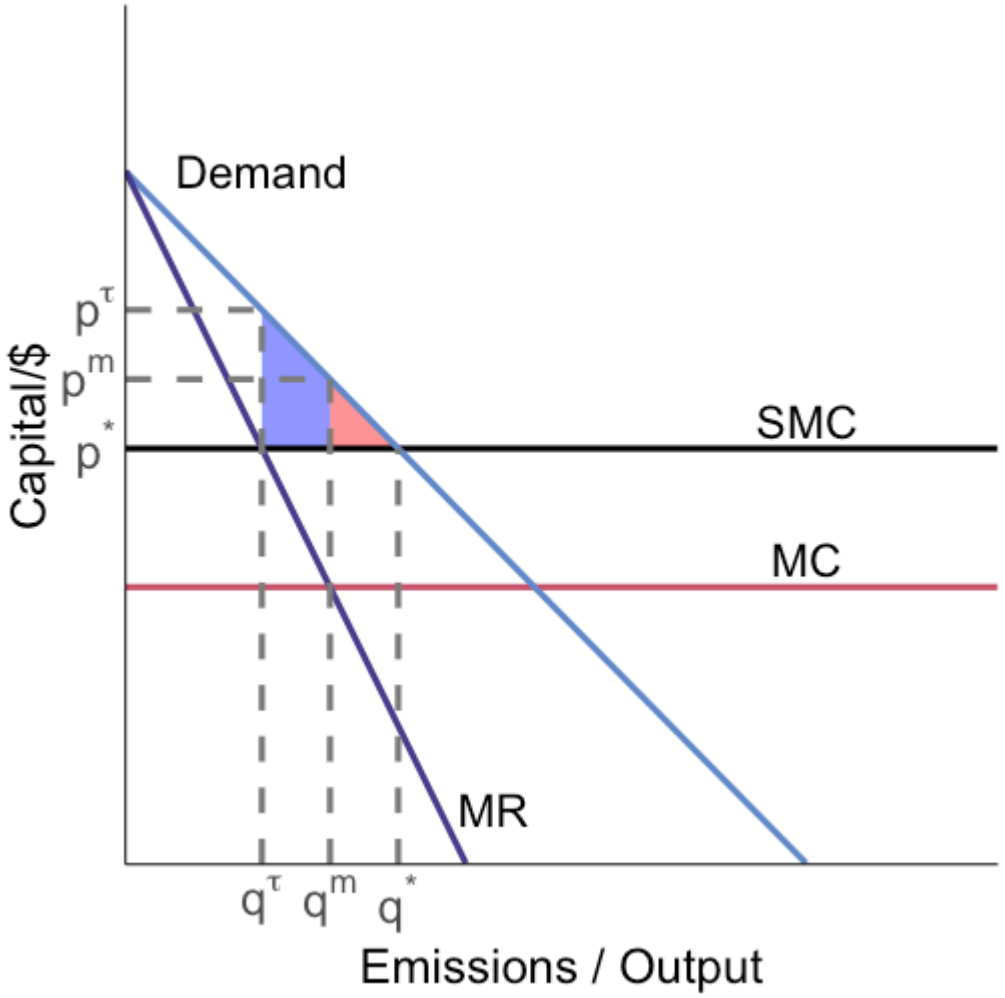


In the absence of regulation, the monopolist maximizes profit where MR = MC: (q^m, p^m)

This results in deadweight loss equal to the **red** area

Now what happens if we set a Pigouvian tax equal to marginal damage?

Monopoly

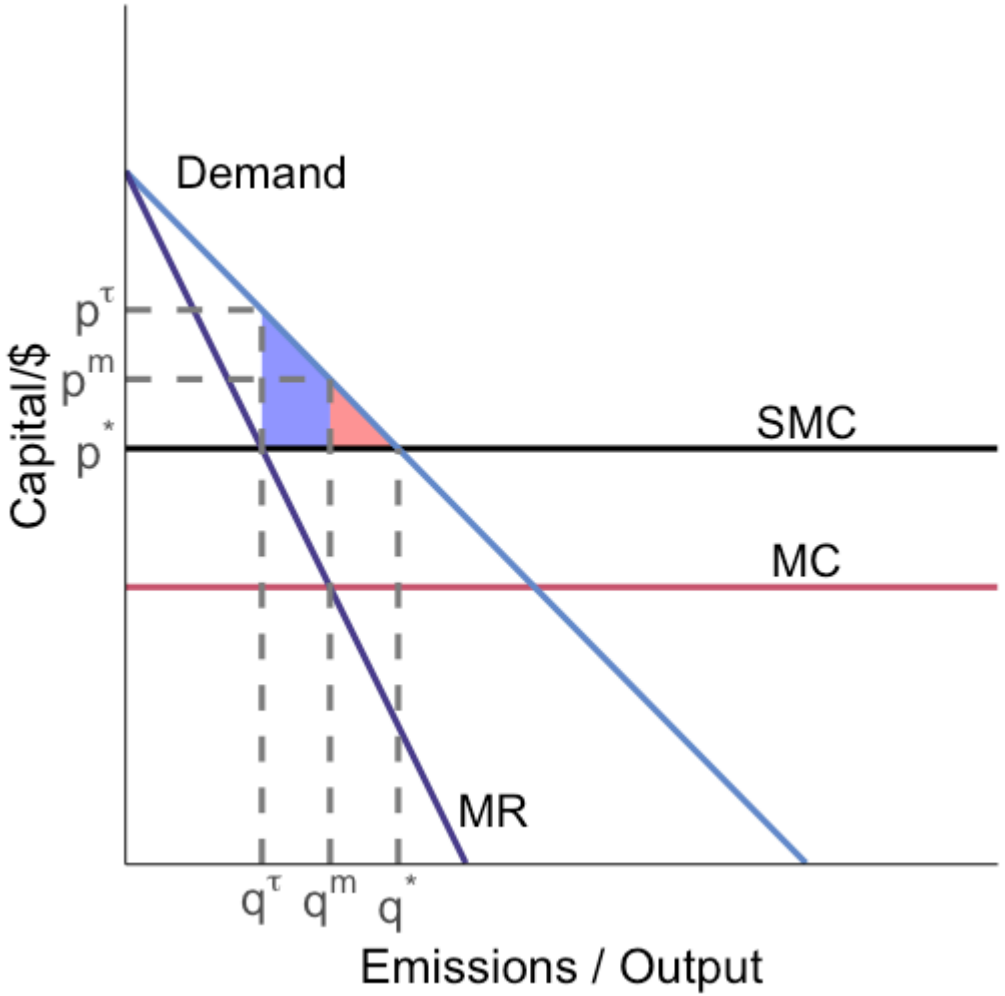


The Pigouvian tax restricts output even more, adding deadweight loss equal to the **blue** area on top of the deadweight loss in the **red** area

The tax actually made us worse off by the blue area!

Why?

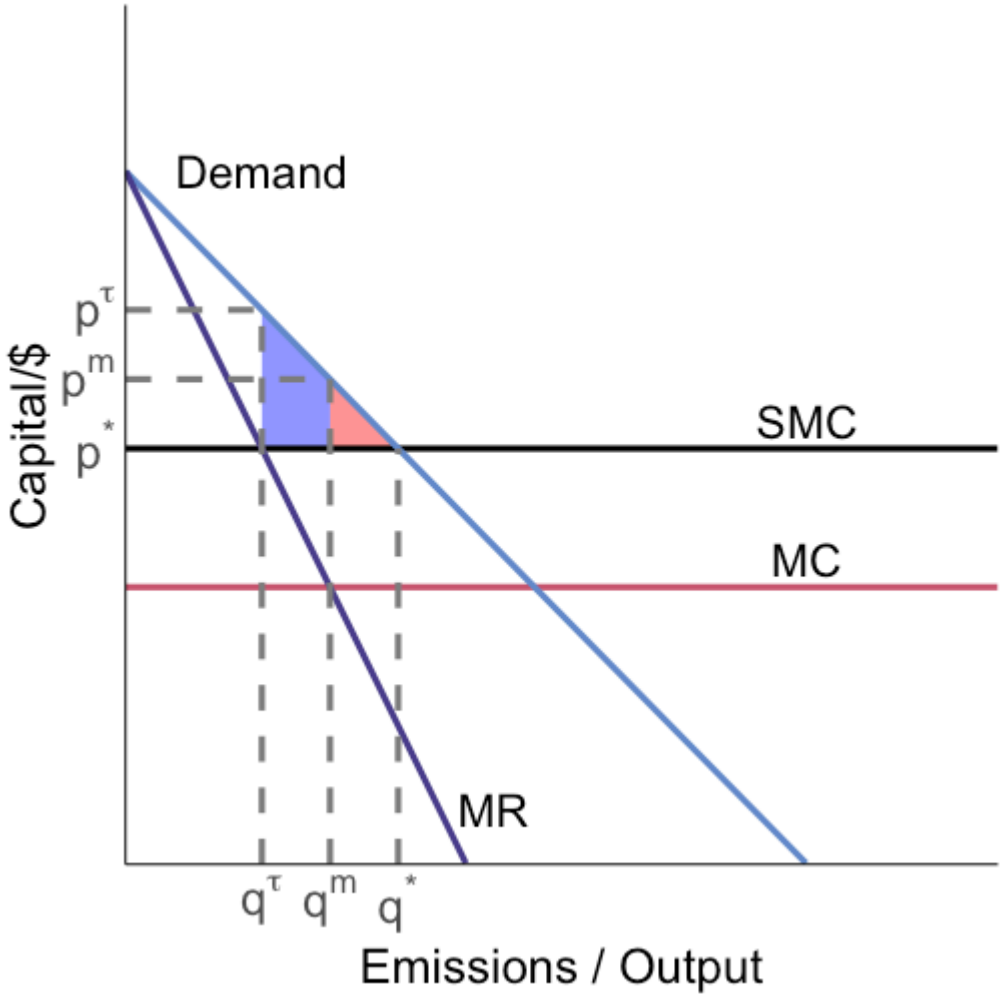
Monopoly



We now have two market failures:

1. Market power
2. Pollution externality

Monopoly

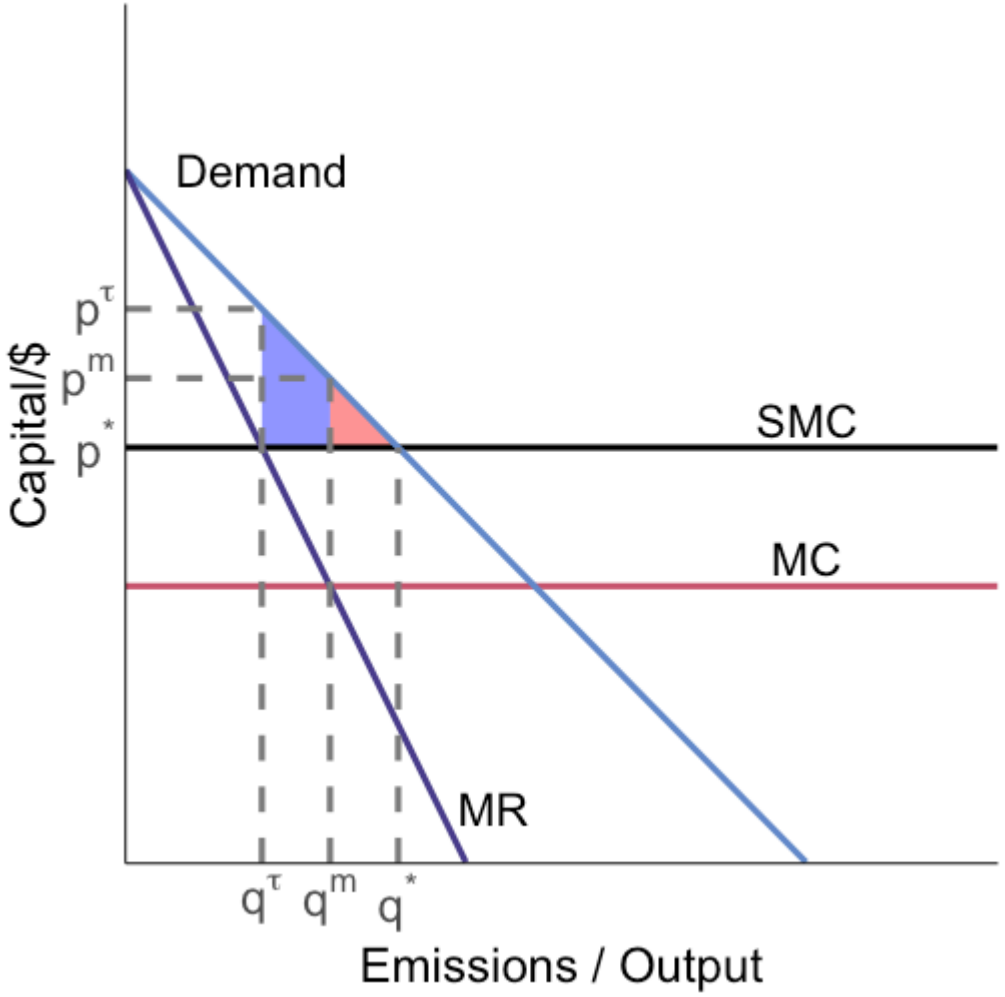


We now have two market failures:

1. Market power
2. Pollution externality

With market power, the unregulated equilibrium quantity is **too low**

Monopoly



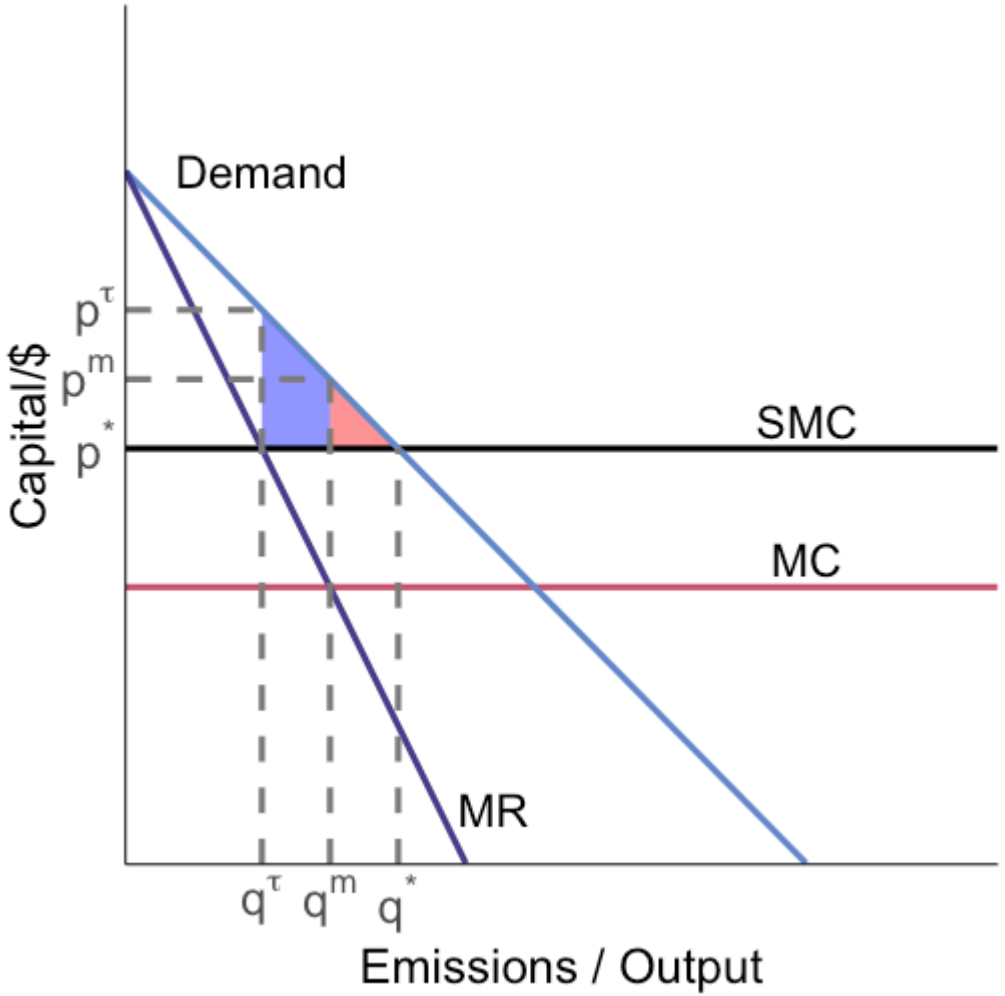
We now have two market failures:

1. Market power
2. Pollution externality

With market power, the unregulated equilibrium quantity is **too low**

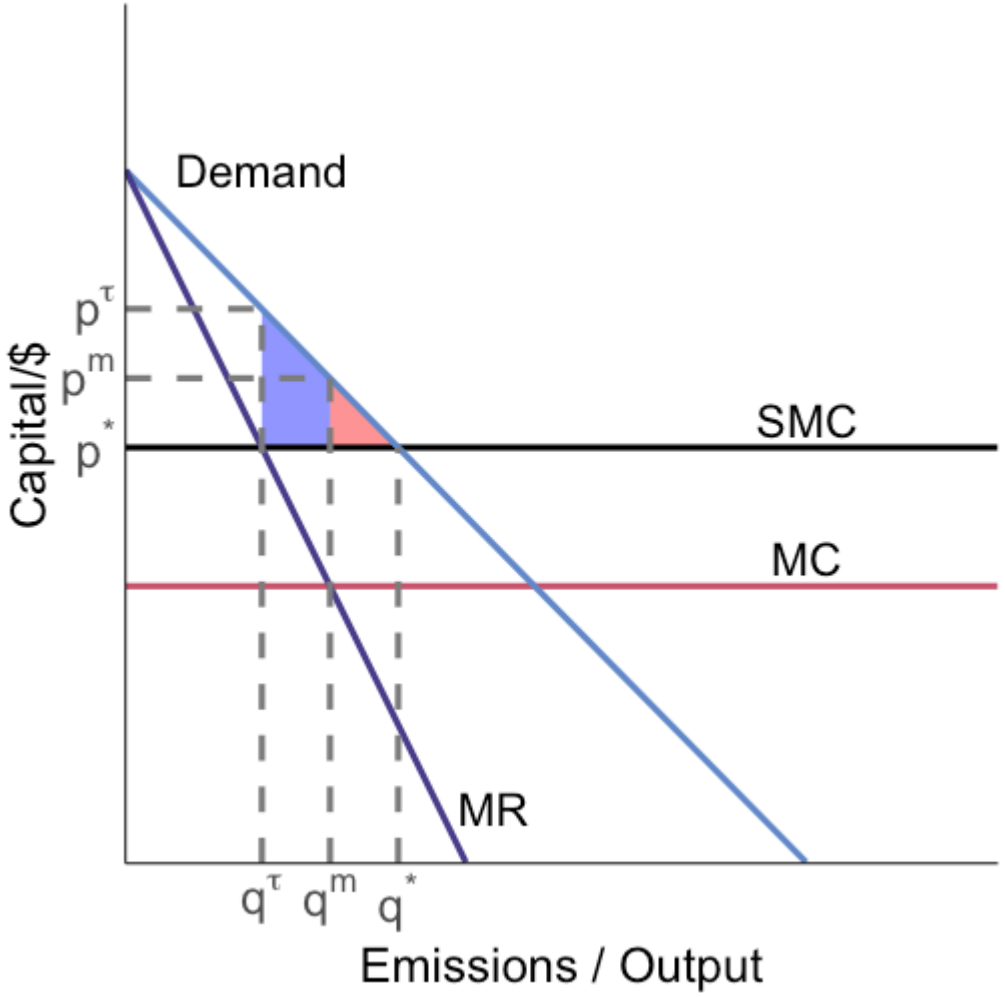
With a pollution externality, the unregulated equilibrium quantity is **too high**

Monopoly



They have opposing forces on quantities, so the market failures offset each other (partially)

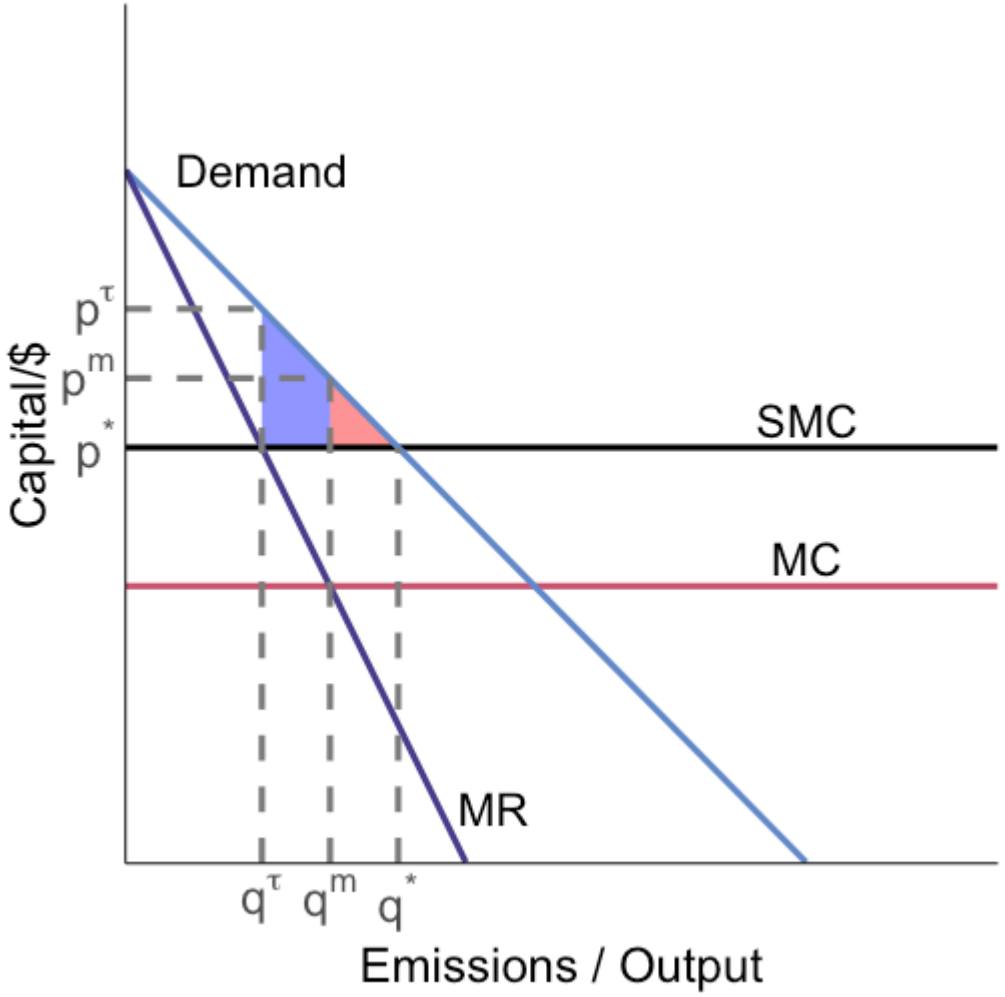
Monopoly



They have opposing forces on quantities, so the market failures offset each other (partially)

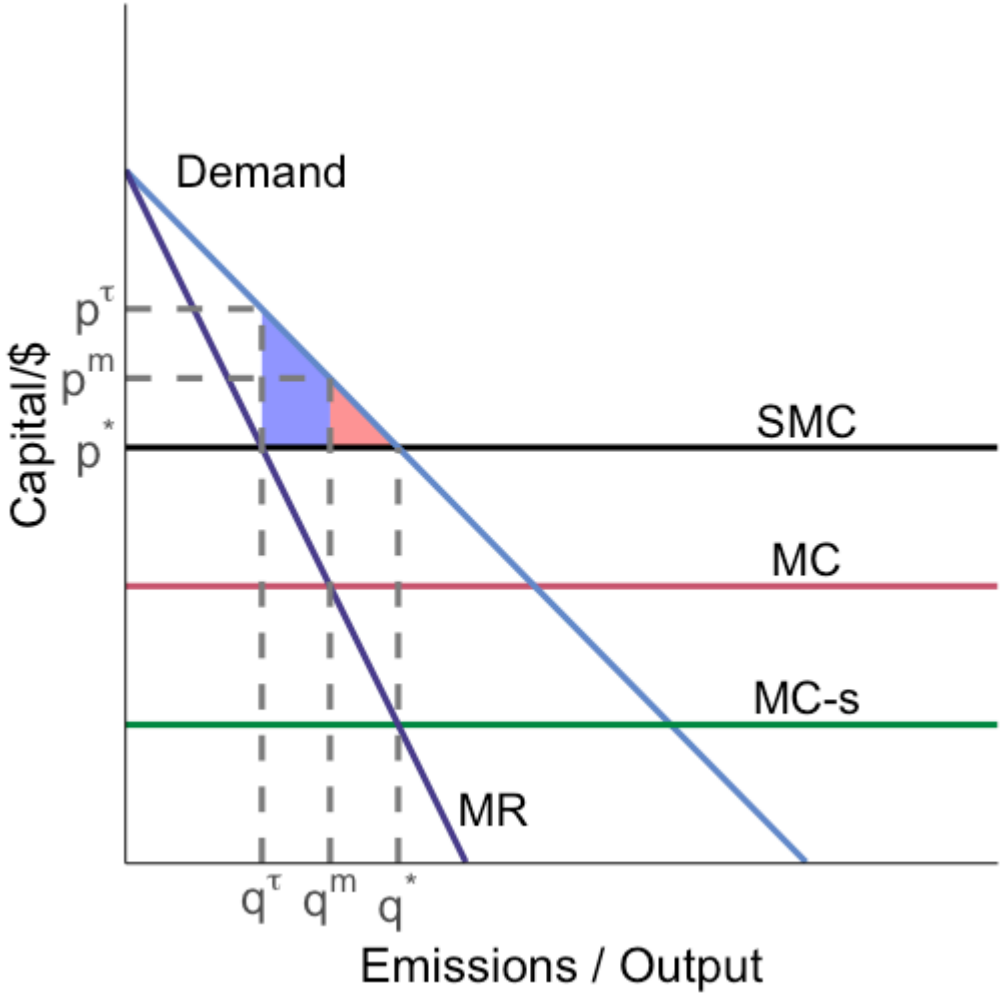
This means that if we fully correct the pollution externality, we no longer get the off-setting benefit and have the full welfare cost of market power

Monopoly



What is the actual optimal thing to do here?

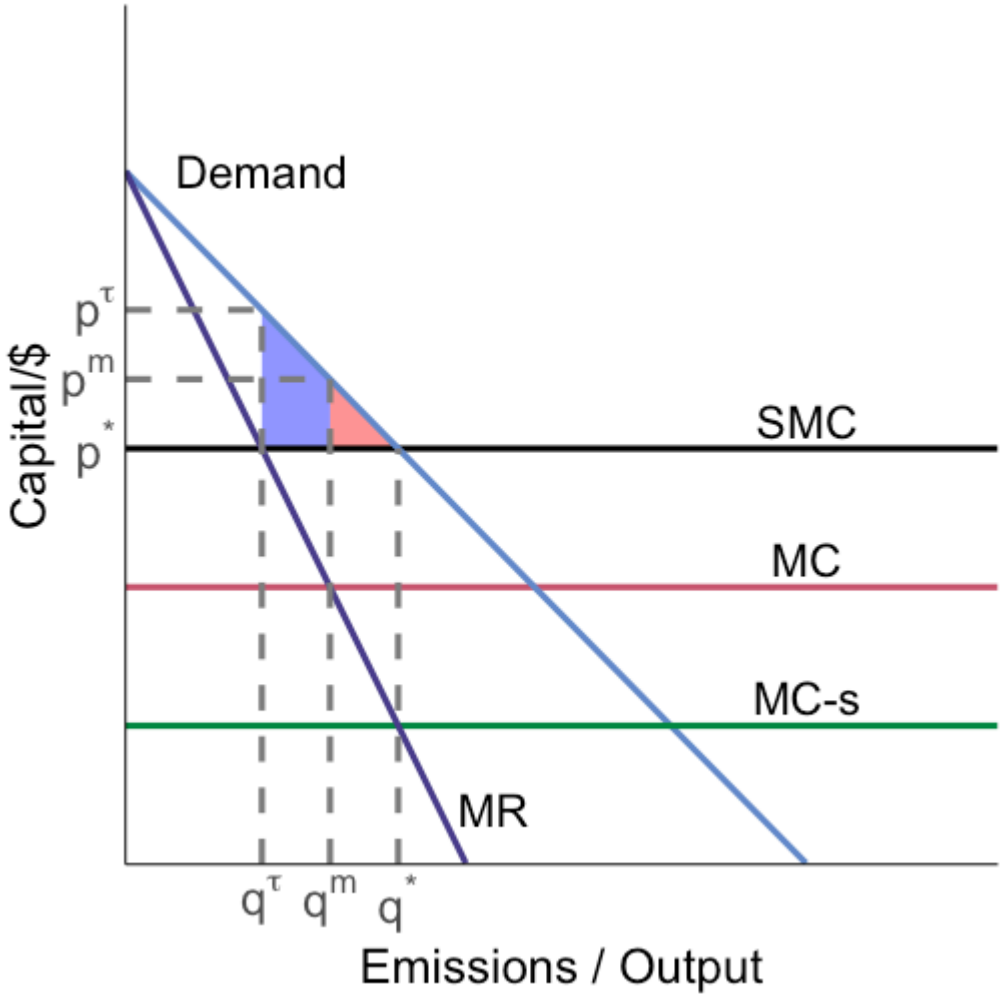
Monopoly



What is the actual optimal thing to do here?

subsidize output at rate s so $MC - s$ crosses MR at q^*

Monopoly



What is the actual optimal thing to do here?

subsidize output at rate s so $MC - s$ crosses MR at q^*

In this example, the market power externality dominates the pollution externality: we need to increase output

Monopoly

What did we learn?

Monopoly

What did we learn?

With multiple market failures we don't necessarily want to **fully** correct for a single market failure

Monopoly

What did we learn?

With multiple market failures we don't necessarily want to **fully** correct for a single market failure

In this example we actually wanted to do the **opposite** of what you likely thought

Monopoly

What did we learn?

With multiple market failures we don't necessarily want to **fully** correct for a single market failure

In this example we actually wanted to do the **opposite** of what you likely thought

You can always draw this example in a different way where you should tax output

Monopoly

What did we learn?

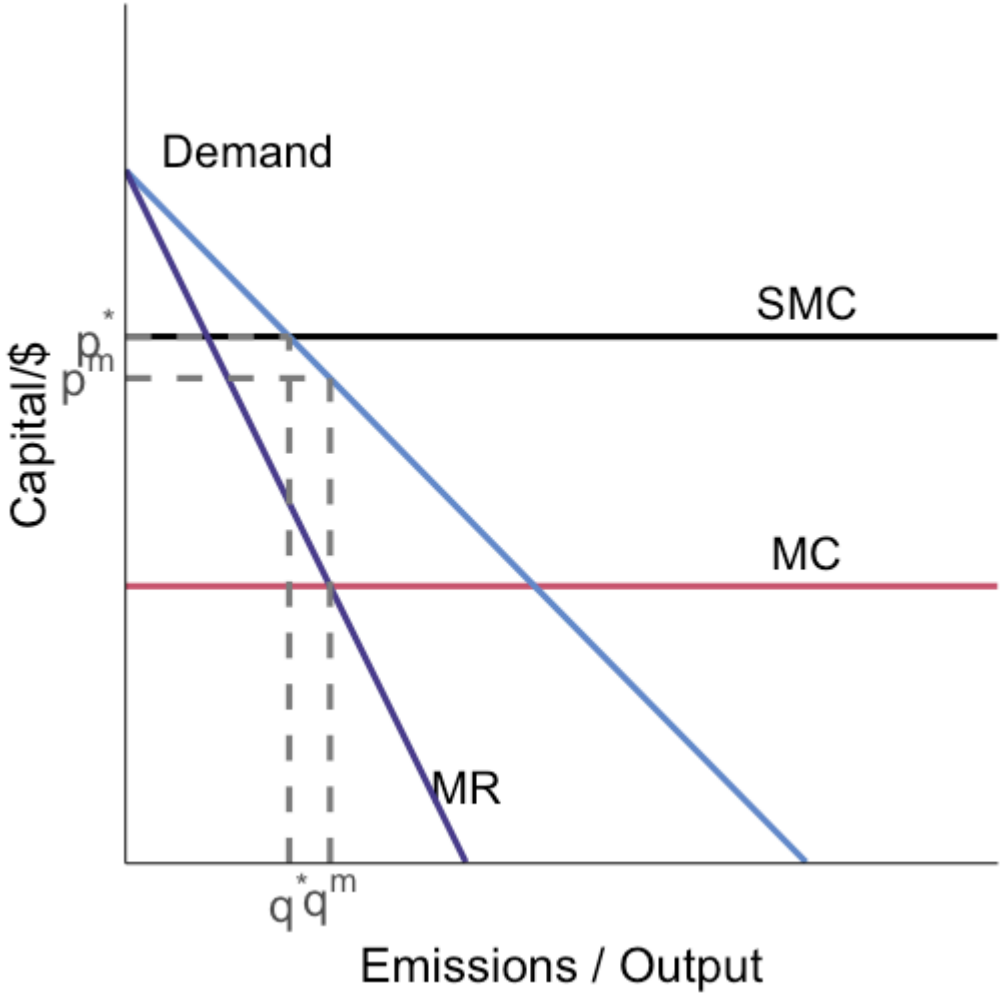
With multiple market failures we don't necessarily want to **fully** correct for a single market failure

In this example we actually wanted to do the **opposite** of what you likely thought

You can always draw this example in a different way where you should tax output

You just need marginal damages to be sufficiently large relative to the market power effect on quantity

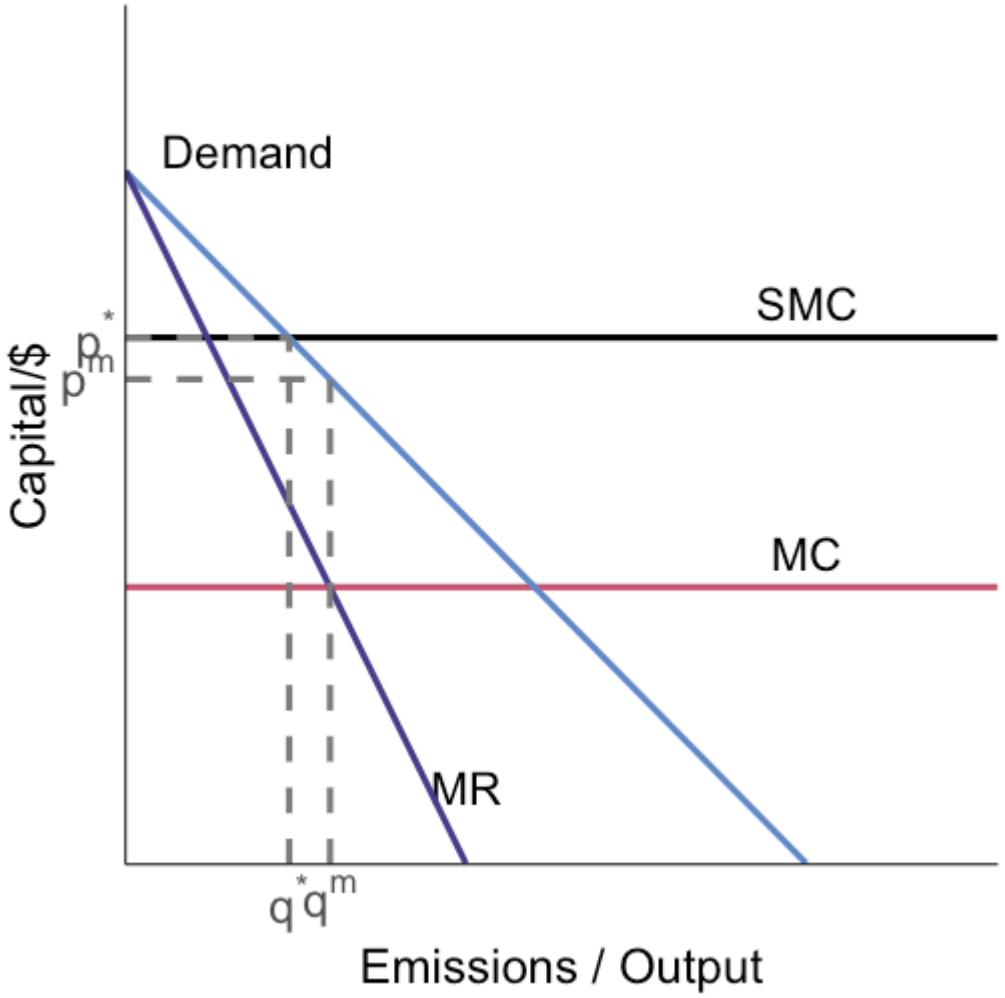
Monopoly



What if marginal damages are larger?

Suppose SMC is high enough that demand and SMC intersect to the **left** of q^m

Monopoly

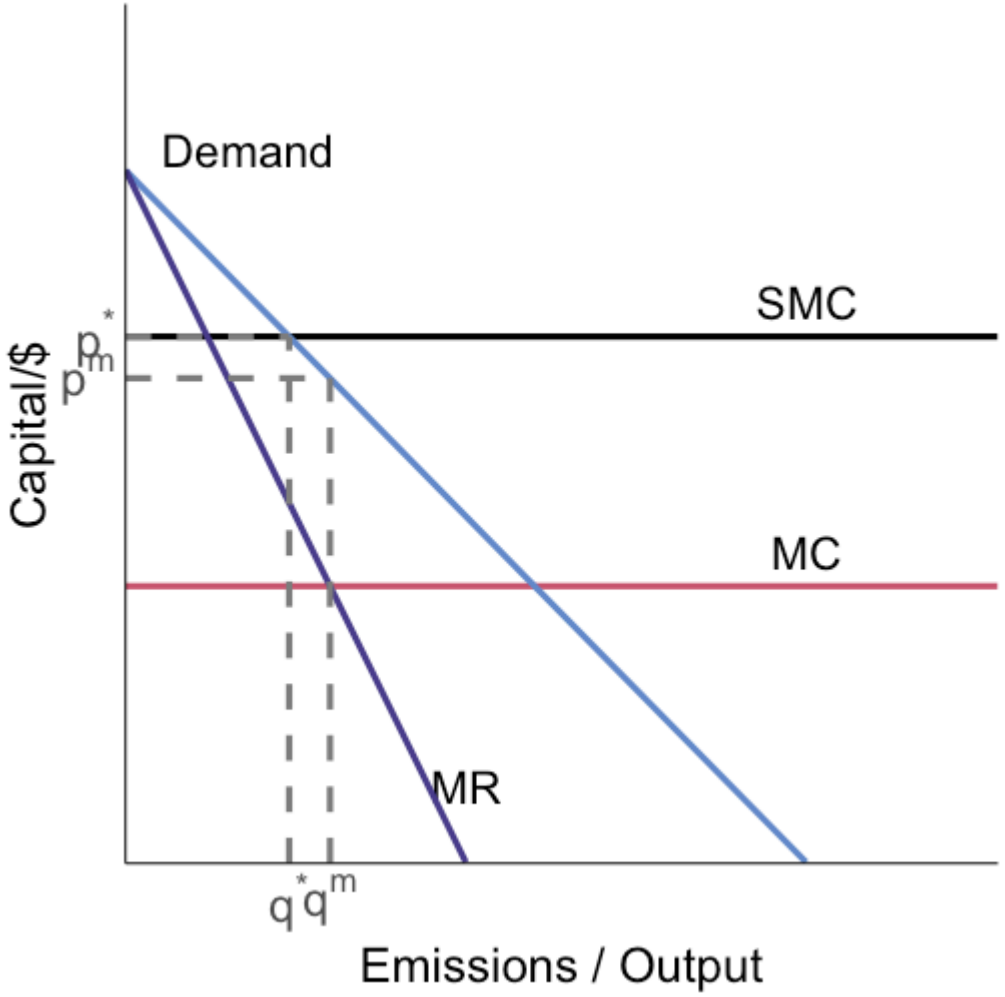


What if marginal damages are larger?

Suppose SMC is high enough that demand and SMC intersect to the **left** of q^m

Then the efficient quantity is lower than monopoly output: $q^* < q^m$

Monopoly

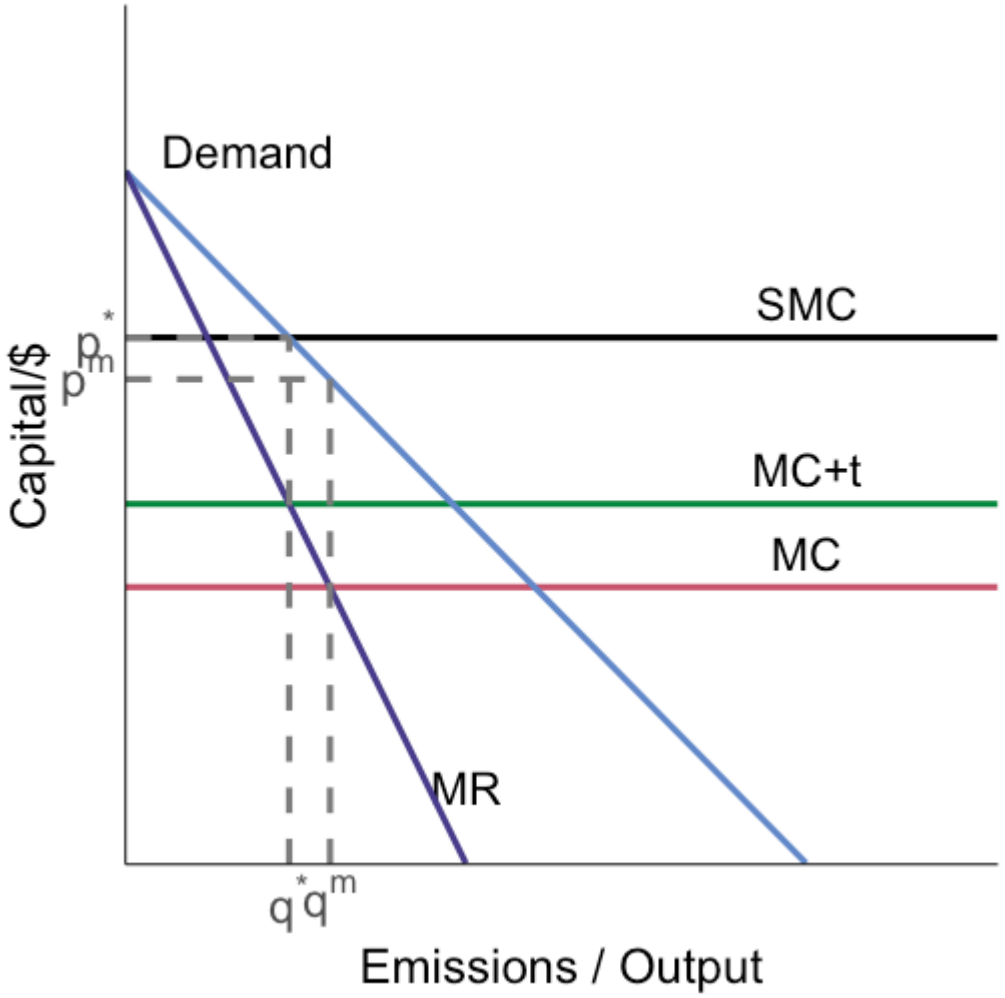


What if marginal damages are larger?

Suppose SMC is high enough that demand and SMC intersect to the **left** of q^m

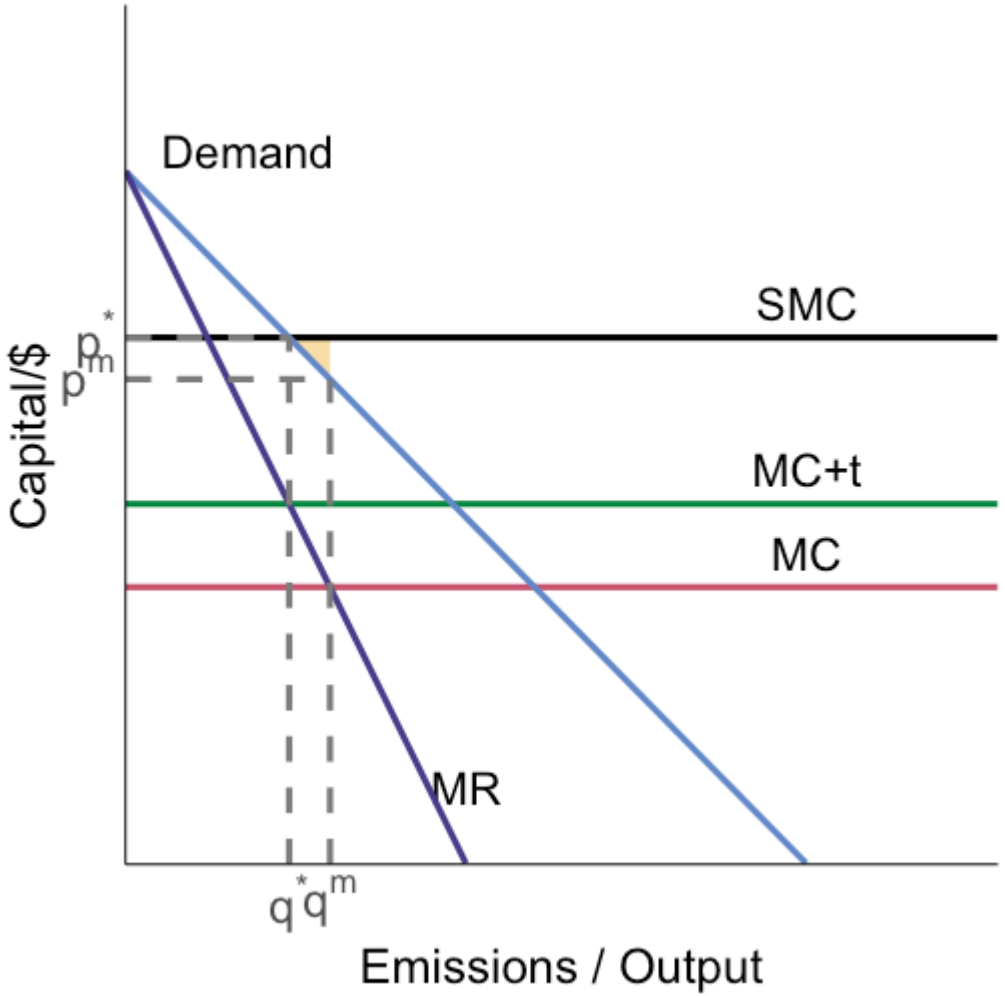
Then the efficient quantity is lower than monopoly output: $q^* < q^m$

Monopoly



Policy implication flips: **tax output** at rate t so $MC + t$ crosses MR at q^*

Monopoly



Policy implication flips: **tax output** at rate t so $MC + t$ crosses MR at q^*

Welfare gain from this correction is the reduction in overproduction between q^m and q^*

Monopoly: more generally

This was a **special case**: marginal damage from output was constant

Monopoly: more generally

This was a **special case**: marginal damage from output was constant

When emissions and output are separate choices, both market failures remain

We can no longer write $E = f(q)$ or marginal damage as a function of output alone

Monopoly: more generally

This was a **special case**: marginal damage from output was constant

When emissions and output are separate choices, both market failures remain

We can no longer write $E = f(q)$ or marginal damage as a function of output alone

A pollution tax/subsidy alone cannot correct both market failures

Monopoly: more generally

Typically we must follow the **Tinbergen rule:** you need as many policy instruments as you have market failures to achieve the efficient outcome

Monopoly: more generally

Typically we must follow the **Tinbergen rule:** you need as many policy instruments as you have market failures to achieve the efficient outcome

What does that mean here?

Monopoly: more generally

Typically we must follow the **Tinbergen rule:** you need as many policy instruments as you have market failures to achieve the efficient outcome

What does that mean here?

We need:

1. Pollution tax

Monopoly: more generally

Typically we must follow the **Tinbergen rule:** you need as many policy instruments as you have market failures to achieve the efficient outcome

What does that mean here?

We need:

1. Pollution tax
2. Output subsidy

The tax incentivizes abatement, the subsidy incentivizes production

Output taxes

Sometimes emission taxes and abatement subsidies are difficult to administer because monitoring is hard

Output taxes

Sometimes emission taxes and abatement subsidies are difficult to administer because monitoring is hard

What can we do instead?

Output taxes

Sometimes emission taxes and abatement subsidies are difficult to administer because monitoring is hard

What can we do instead?

We can tax the output of the production process

Output taxes

Sometimes emission taxes and abatement subsidies are difficult to administer because monitoring is hard

What can we do instead?

We can tax the output of the production process

e.g. tax on electricity, cement

Output taxes

Sometimes emission taxes and abatement subsidies are difficult to administer because monitoring is hard

What can we do instead?

We can tax the output of the production process

e.g. tax on electricity, cement

Will this be efficient?

Output taxes

Sometimes emission taxes and abatement subsidies are difficult to administer because monitoring is hard

What can we do instead?

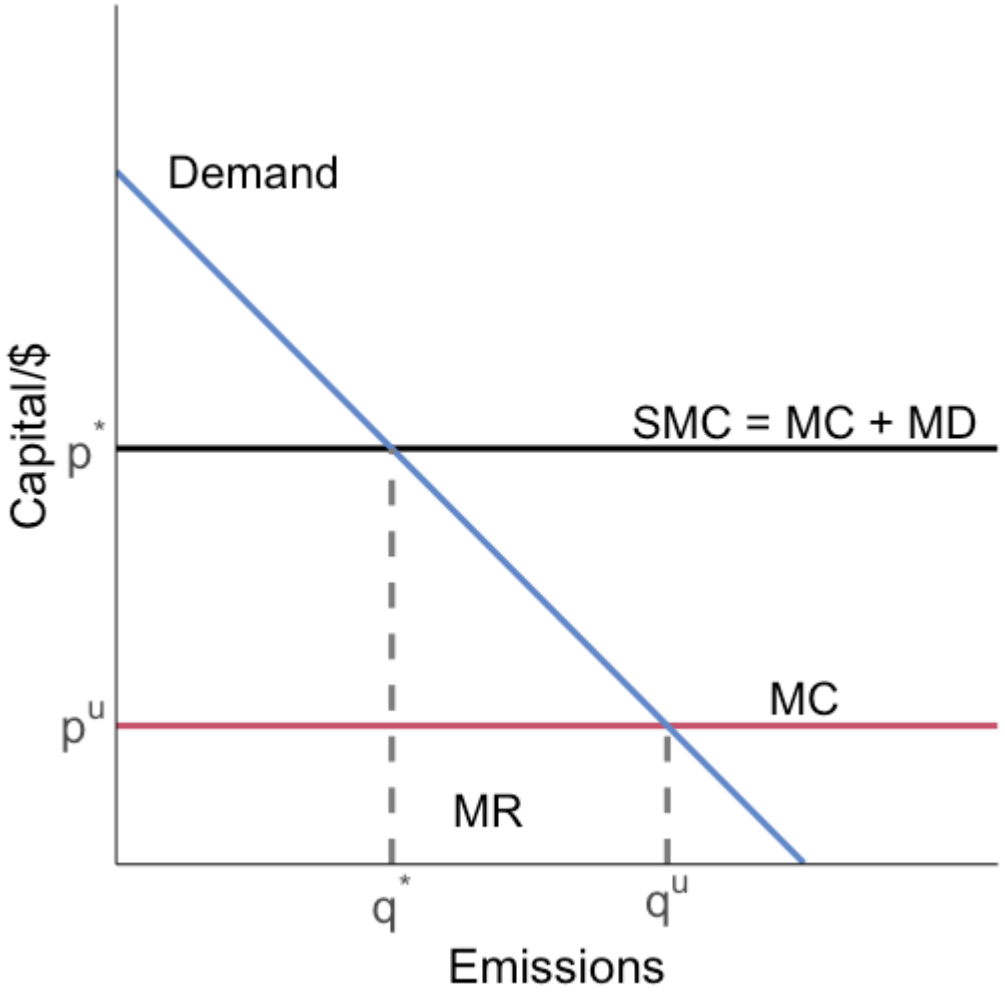
We can tax the output of the production process

e.g. tax on electricity, cement

Will this be efficient?

If so, what assumptions do we need about the production process?

Output taxes



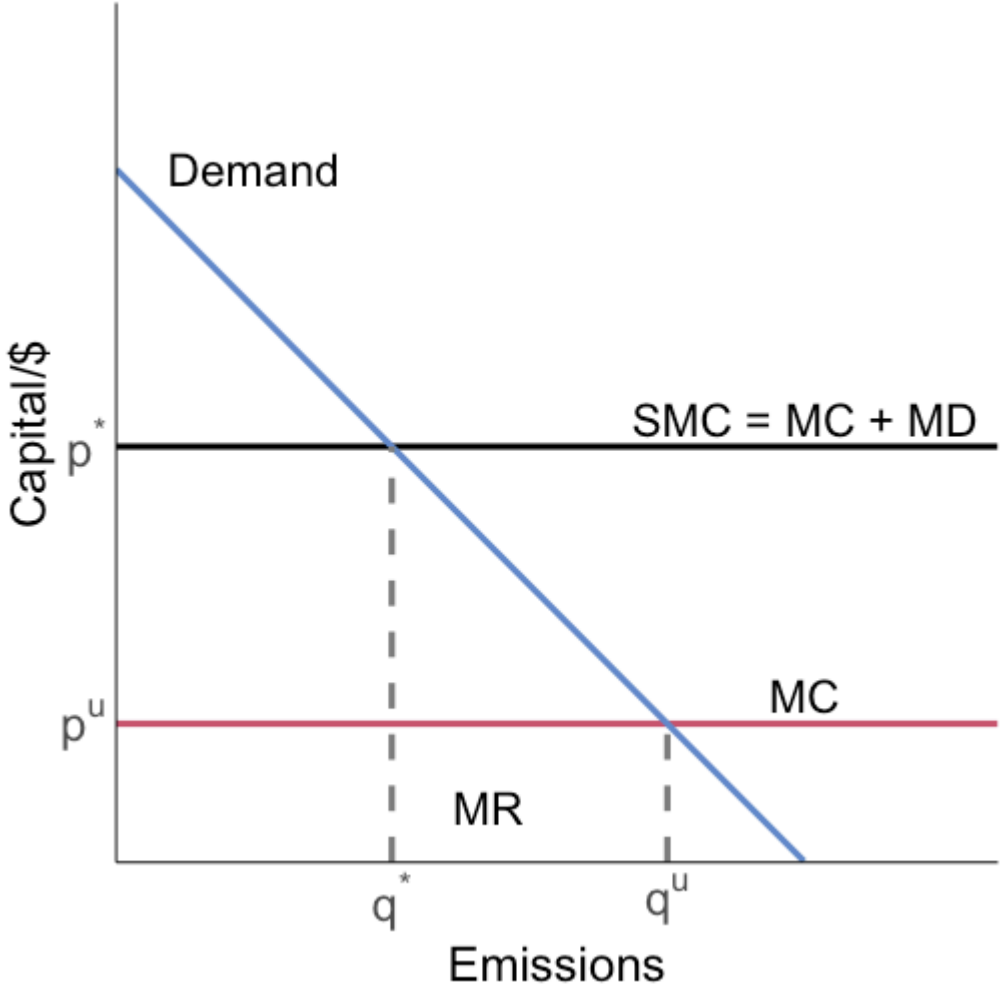
Assume emissions are proportional to output

And MD is constant

The firm chooses to produce/emit at q^u in the unregulated equilibrium

If we tax output equal to MD we can achieve the socially optimal allocation q^*

Output taxes



An output tax can be efficient, **if we assume that emissions are proportional to output**

Now let's break the link between output and emissions by writing down a slightly more complicated model where the firm chooses emissions and output separately

Output taxes, part two

Here's our model:

- Production uses labor L and energy E: $Q = f(L, E)$
- Labor costs w per unit and energy costs r per unit
- The firm receives a price p per unit of output
- 1 unit of energy causes 1 unit of emissions, with marginal damage d per unit of emissions

Output taxes, part two

Here's our model:

- Production uses labor L and energy E: $Q = f(L, E)$
- Labor costs w per unit and energy costs r per unit
- The firm receives a price p per unit of output
- 1 unit of energy causes 1 unit of emissions, with marginal damage d per unit of emissions

The key point is that the firm can adjust L and E separately

Output taxes, part two

Here's our model:

- Production uses labor L and energy E: $Q = f(L, E)$
- Labor costs w per unit and energy costs r per unit
- The firm receives a price p per unit of output
- 1 unit of energy causes 1 unit of emissions, with marginal damage d per unit of emissions

The key point is that the firm can adjust L and E separately

What does an output tax τ_o do versus a regular emission tax τ_e?

Output taxes, part two

The regulator wants the firm to internalize its social costs:

$$\max_{L,E} pf(L,E) - wL - rE - dE$$

Output taxes, part two

The regulator wants the firm to internalize its social costs:

$$\max_{L,E} pf(L,E) - wL - rE - dE$$

The first-order conditions for a socially efficient allocation of L and E are:

$$p\frac{\partial f(L,E)}{\partial L} = w \qquad p\frac{\partial f(L,E)}{\partial E} = r + d$$

Output taxes, part two

The regulator wants the firm to internalize its social costs:

$$\max_{L,E} pf(L,E) - wL - rE - dE$$

The first-order conditions for a socially efficient allocation of L and E are:

$$p\frac{\partial f(L,E)}{\partial L} = w \qquad p\frac{\partial f(L,E)}{\partial E} = r + d$$

For a social optimum, labor is priced at its private cost, but emissions must face the extra damage term d

Output taxes, part two

The firm's problem for the output tax is:

$$\max_{L,E}(p-\tau_o)f(L,E)-wL-rE$$

Output taxes, part two

The firm's problem for the output tax is:

$$\max_{L,E}(p-\tau_o)f(L,E)-wL-rE$$

The firm's profit-maximizing choices are given by the first-order conditions:

$$(p-\tau_o)\frac{\partial f(L,E)}{\partial L}=w \qquad (p-\tau_o)\frac{\partial f(L,E)}{\partial E}=r$$

Output taxes, part two

The firm's problem for the output tax is:

$$\max_{L,E}(p-\tau_o)f(L,E)-wL-rE$$

The firm's profit-maximizing choices are given by the first-order conditions:

$$(p-\tau_o)\frac{\partial f(L,E)}{\partial L}=w \qquad (p-\tau_o)\frac{\partial f(L,E)}{\partial E}=r$$

What do these conditions mean?

Output taxes, part two

The output tax reduces the value of the marginal product of **both** inputs

Output taxes, part two

The output tax reduces the value of the marginal product of **both** inputs

Rearranging the labor condition gives:

$$p\frac{\partial f(L,E)}{\partial L} = w + \tau_o \frac{\partial f(L,E)}{\partial L}$$

Output taxes, part two

The output tax reduces the value of the marginal product of **both** inputs

Rearranging the labor condition gives:

$$p\frac{\partial f(L,E)}{\partial L} = w + \tau_o \frac{\partial f(L,E)}{\partial L}$$

So the output tax implicitly taxes clean labor too, which is **not efficient**

Output taxes, part two

How does this compare to a pure emission tax?

Output taxes, part two

How does this compare to a pure emission tax?

The firm's problem when facing an emission tax is:

$$\max_{L,E} pf(L,E) - wL - (r + \tau_e)E$$

Output taxes, part two

How does this compare to a pure emission tax?

The firm's problem when facing an emission tax is:

$$\max_{L,E} pf(L,E) - wL - (r+\tau_e)E$$

The firm's profit-maximizing choices are given by the first-order conditions:

$$p\frac{\partial f(L,E)}{\partial L} = w \qquad p\frac{\partial f(L,E)}{\partial E} = r + \tau_e$$

Output taxes, part two

How does this compare to a pure emission tax?

The firm's problem when facing an emission tax is:

$$\max_{L,E} pf(L,E) - wL - (r+\tau_e)E$$

The firm's profit-maximizing choices are given by the first-order conditions:

$$p\frac{\partial f(L,E)}{\partial L} = w \qquad p\frac{\partial f(L,E)}{\partial E} = r + \tau_e$$

A tax of $\tau_e = d$ can achieve the efficient allocation!

Output taxes takeaways

If emissions are perfectly determined by output, we can use output taxes to achieve the efficient outcome

Output taxes takeaways

If emissions are perfectly determined by output, we can use output taxes to achieve the efficient outcome

If emissions can be chosen separately from outcome by the firm, this is no longer true

Output taxes takeaways

If emissions are perfectly determined by output, we can use output taxes to achieve the efficient outcome

If emissions can be chosen separately from outcome by the firm, this is no longer true

In this case an output tax incorrectly taxes our clean inputs

Intensity standards

Another common second-best policy is an **intensity standard**

Intensity standards

Another common second-best policy is an **intensity standard**

Instead of pricing each unit of emissions directly, the regulator sets an emissions rate per unit of output

Intensity standards

Another common second-best policy is an **intensity standard**

Instead of pricing each unit of emissions directly, the regulator sets an emissions rate per unit of output

These policies are common when direct emissions monitoring is difficult but output composition is observable

Intensity standards: RPS example

What is a Renewables Portfolio Standard (RPS)?

aka Renewable Energy/Electricity Standard (RES)

Renewables Portfolio Standard	A requirement on retail electric suppliers… To supply a minimum percentage or amount of their retail load… With eligible sources of renewable energy
Typically	Backed with penalties of some form
Often	Accompanied by a tradable renewable energy certificate (REC) program to facilitate compliance
Never	Designed the same in any two states

This report covers U.S. state RPS policies. It does <u>not</u> cover:

- Voluntary renewable electricity goals
- Broader clean energy requirements without a renewables-specific component (briefly discussed in a side-bar)
- RPS policies outside of the United States or in U.S. territories

Intensity standards: the constraint

Let H be the dirtier product and L the cleaner product

Intensity standards: the constraint

Let H be the dirtier product and L the cleaner product

An intensity standard requires:

$$\frac{\beta_H Q_H + \beta_L Q_L}{Q_H + Q_L} \leq \sigma$$

Intensity standards: the constraint

Let H be the dirtier product and L the cleaner product

An intensity standard requires:

$$\frac{\beta_H Q_H + \beta_L Q_L}{Q_H + Q_L} \leq \sigma$$

- β_i: emissions per unit of product i
- σ: maximum average emissions per unit of output
- Assume $\beta_H > \sigma > \beta_L$: H is above the standard, L below

Intensity standards: the constraint

Let H be the dirtier product and L the cleaner product

An intensity standard requires:

$$\frac{\beta_H Q_H + \beta_L Q_L}{Q_H + Q_L} \leq \sigma$$

- β_i: emissions per unit of product i
- σ: maximum average emissions per unit of output
- Assume $\beta_H > \sigma > \beta_L$: H is above the standard, L below

The standard limits average emissions; total emissions can still rise

Intensity standards: firm's problem

Suppose the firm takes output prices p_H and p_L as given

Intensity standards: firm's problem

Suppose the firm takes output prices p_H and p_L as given

Its private problem is:

$$\max_{Q_H, Q_L} \pi(Q_H, Q_L) = p_H Q_H + p_L Q_L - C_H(Q_H) - C_L(Q_L)$$

Intensity standards: firm's problem

Suppose the firm takes output prices p_H and p_L as given

Its private problem is:

$$\max_{Q_H, Q_L} \pi(Q_H, Q_L) = p_H Q_H + p_L Q_L - C_H(Q_H) - C_L(Q_L)$$

subject to the intensity standard:

$$\frac{\beta_H Q_H + \beta_L Q_L}{Q_H + Q_L} \leq \sigma$$

Intensity standards: Lagrangian setup

Rewrite the constraint as:

$$\sigma(Q_H + Q_L) - \beta_H Q_H - \beta_L Q_L \geq 0$$

Intensity standards: Lagrangian setup

Rewrite the constraint as:

$$\sigma(Q_H + Q_L) - \beta_H Q_H - \beta_L Q_L \geq 0$$

The Lagrangian is:

$$\mathcal{L} = p_H Q_H + p_L Q_L - C_H(Q_H) - C_L(Q_L) + \lambda \left[\sigma(Q_H + Q_L) - \beta_H Q_H - \beta_L Q_L\right]$$

Intensity standards: Lagrangian setup

Rewrite the constraint as:

$$\sigma(Q_H + Q_L) - \beta_H Q_H - \beta_L Q_L \geq 0$$

The Lagrangian is:

$$\mathcal{L} = p_H Q_H + p_L Q_L - C_H(Q_H) - C_L(Q_L) + \lambda \left[\sigma(Q_H + Q_L) - \beta_H Q_H - \beta_L Q_L\right]$$

with $\lambda \geq 0$; if the standard binds, λ is the shadow value of relaxing it

Intensity standards: firm incentives

Taking first-order conditions from the Lagrangian gives:

$$p_i = MC_i(Q_i) + \lambda(\beta_i - \sigma), \quad i \in \{H, L\}$$

Intensity standards: firm incentives

Taking first-order conditions from the Lagrangian gives:

$$p_i = MC_i(Q_i) + \lambda(\beta_i - \sigma), \quad i \in \{H, L\}$$

If $\beta_i > \sigma$, the policy creates an implicit **tax** on that output

Intensity standards: firm incentives

Taking first-order conditions from the Lagrangian gives:

$$p_i = MC_i(Q_i) + \lambda(\beta_i - \sigma), \quad i \in \{H, L\}$$

If $\beta_i > \sigma$, the policy creates an implicit **tax** on that output

If $\beta_i < \sigma$, the policy creates an implicit **subsidy** on that output

Intensity standards: equivalent policy mix

Rewrite the wedge term as:

$$\lambda(\beta_i - \sigma) = \underbrace{\lambda\beta_i}_{\text{emissions tax piece}} - \underbrace{\lambda\sigma}_{\text{output subsidy piece}}$$

Intensity standards: equivalent policy mix

Rewrite the wedge term as:

$$\lambda(\beta_i - \sigma) = \underbrace{\lambda\beta_i}_{\text{emissions tax piece}} - \underbrace{\lambda\sigma}_{\text{output subsidy piece}}$$

An **emissions tax** of λ per unit of emissions costs $\lambda\beta_i$ per unit of output, so dirtier goods pay more

Intensity standards: equivalent policy mix

Rewrite the wedge term as:

$$\lambda(\beta_i - \sigma) = \underbrace{\lambda\beta_i}_{\text{emissions tax piece}} - \underbrace{\lambda\sigma}_{\text{output subsidy piece}}$$

An **emissions tax** of λ per unit of emissions costs $\lambda\beta_i$ per unit of output, so dirtier goods pay more

An **output subsidy** of $\lambda\sigma$ per unit is the same for every good, regardless of β_i

Intensity standards: cleaner is not clean

Suppose $\beta_H = 1.0$, $\beta_L = 0.4$, and $\sigma = 0.6$

Intensity standards: cleaner is not clean

Suppose $\beta_H = 1.0$, $\beta_L = 0.4$, and $\sigma = 0.6$

Then the policy wedges are:

- for H: $\lambda(1.0 - 0.6) = +0.4\lambda$ (implicit tax)
- for L: $\lambda(0.4 - 0.6) = -0.2\lambda$ (implicit subsidy)

Intensity standards: cleaner is not clean

Suppose $\beta_H = 1.0$, $\beta_L = 0.4$, and $\sigma = 0.6$

Then the policy wedges are:

- for H: $\lambda(1.0 - 0.6) = +0.4\lambda$ (implicit tax)
- for L: $\lambda(0.4 - 0.6) = -0.2\lambda$ (implicit subsidy)

The lower-intensity good still pollutes, but it can get subsidized anyway

Intensity standards: cleaner is not clean

Suppose $\beta_H = 1.0$, $\beta_L = 0.4$, and $\sigma = 0.6$

Then the policy wedges are:

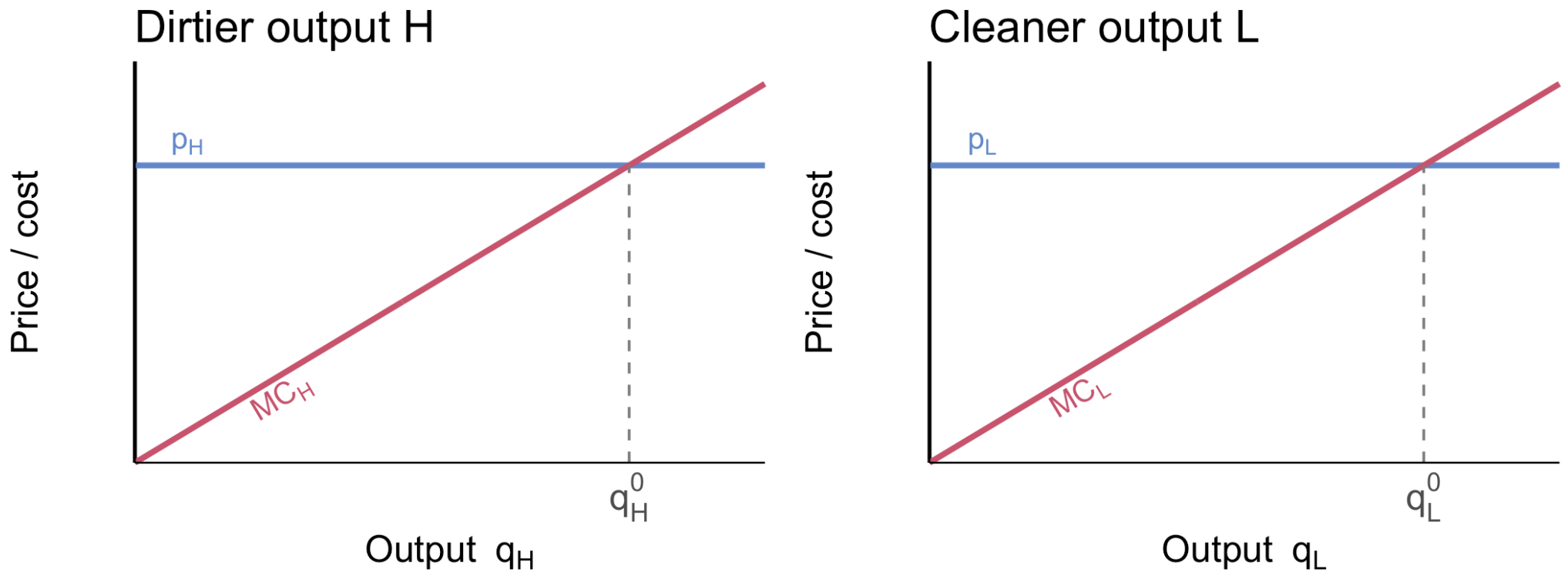
- for H: $\lambda(1.0 - 0.6) = +0.4\lambda$ (implicit tax)
- for L: $\lambda(0.4 - 0.6) = -0.2\lambda$ (implicit subsidy)

The lower-intensity good still pollutes, but it can get subsidized anyway

That is the core problem: we generally do **not** want to subsidize positive emissions

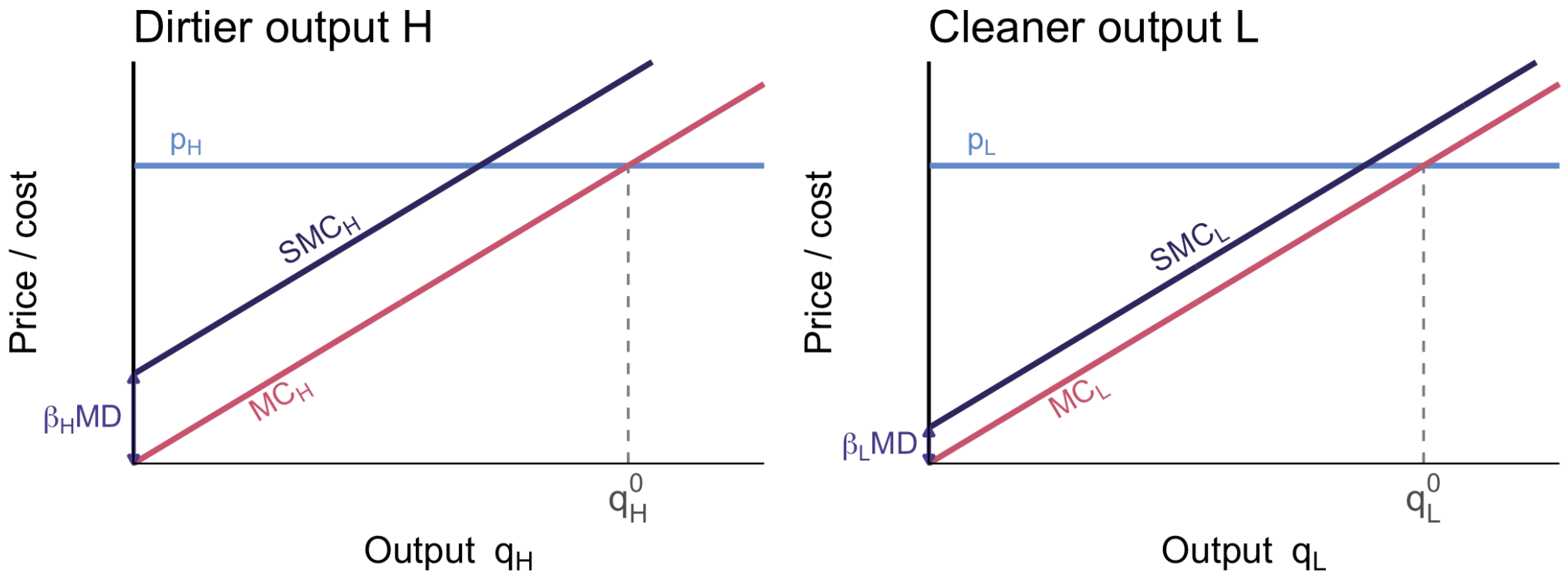
Intensity standards: baseline DWL

Firms take prices as given and choose output where $p_i = MC_i$



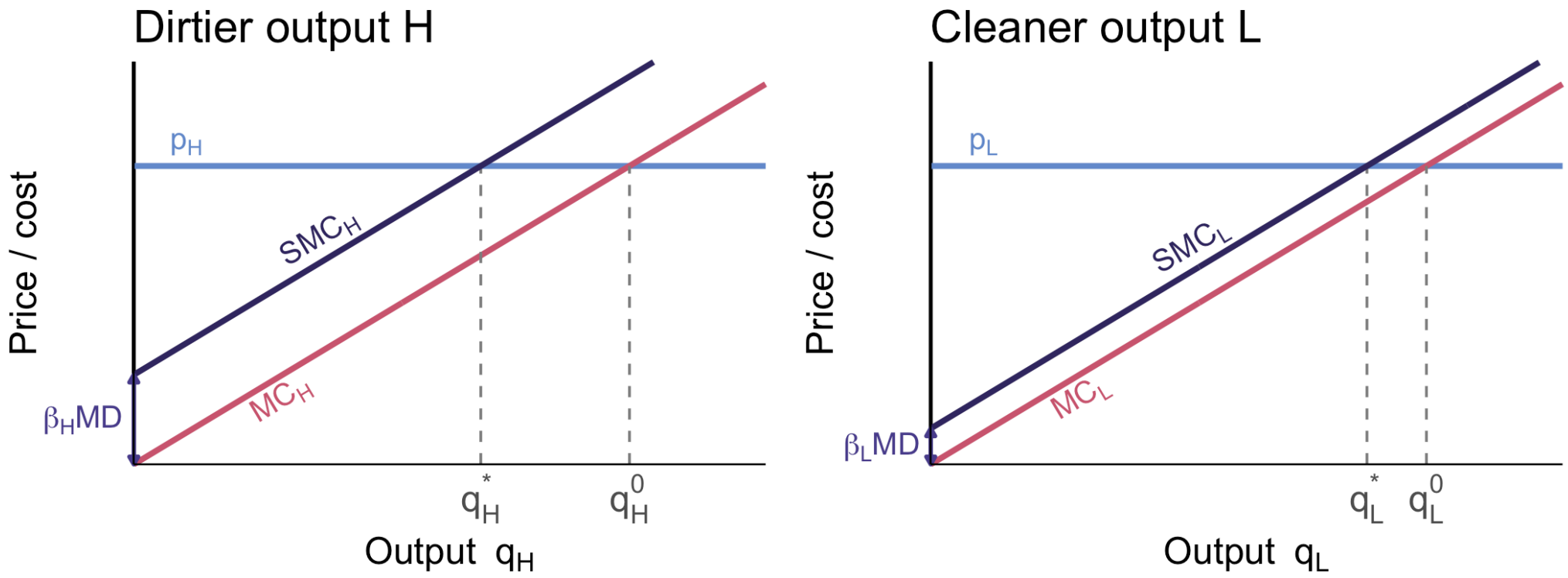
Intensity standards: baseline DWL

Social marginal cost adds emissions damages: $SMC_i = MC_i + \beta_i MD$



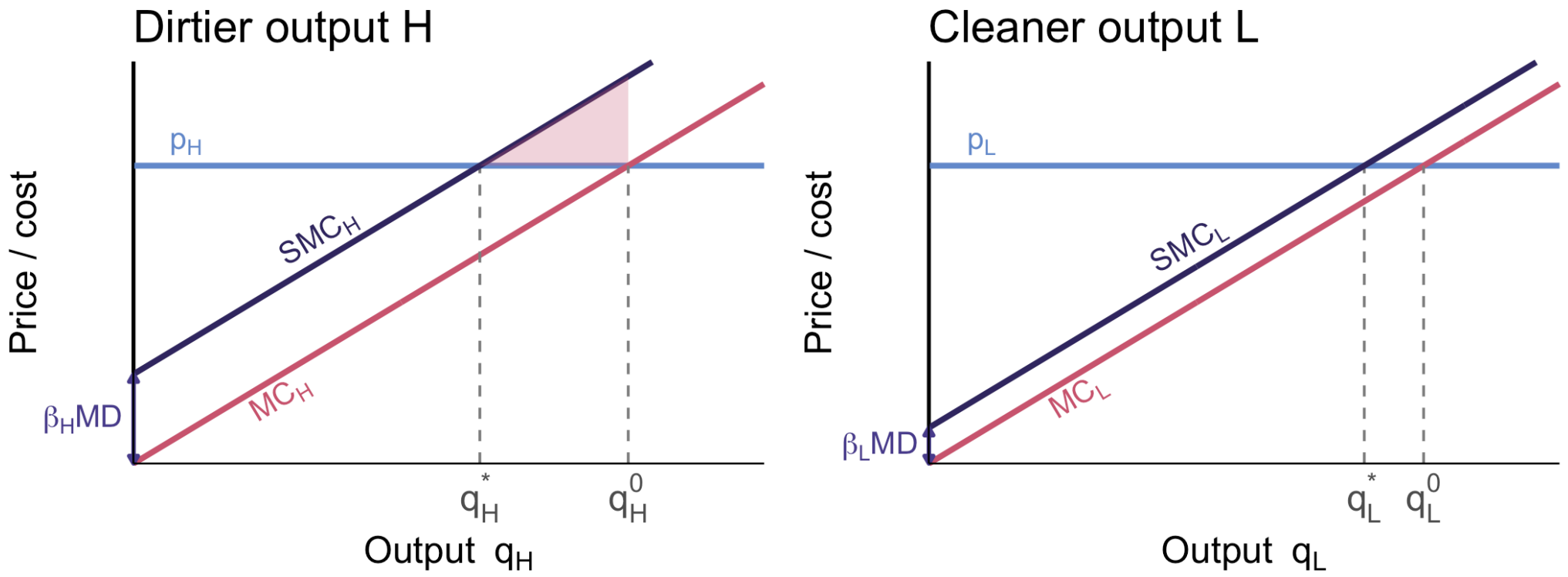
Intensity standards: baseline DWL

Efficient output is where $p_i = SMC_i$; both markets overproduce



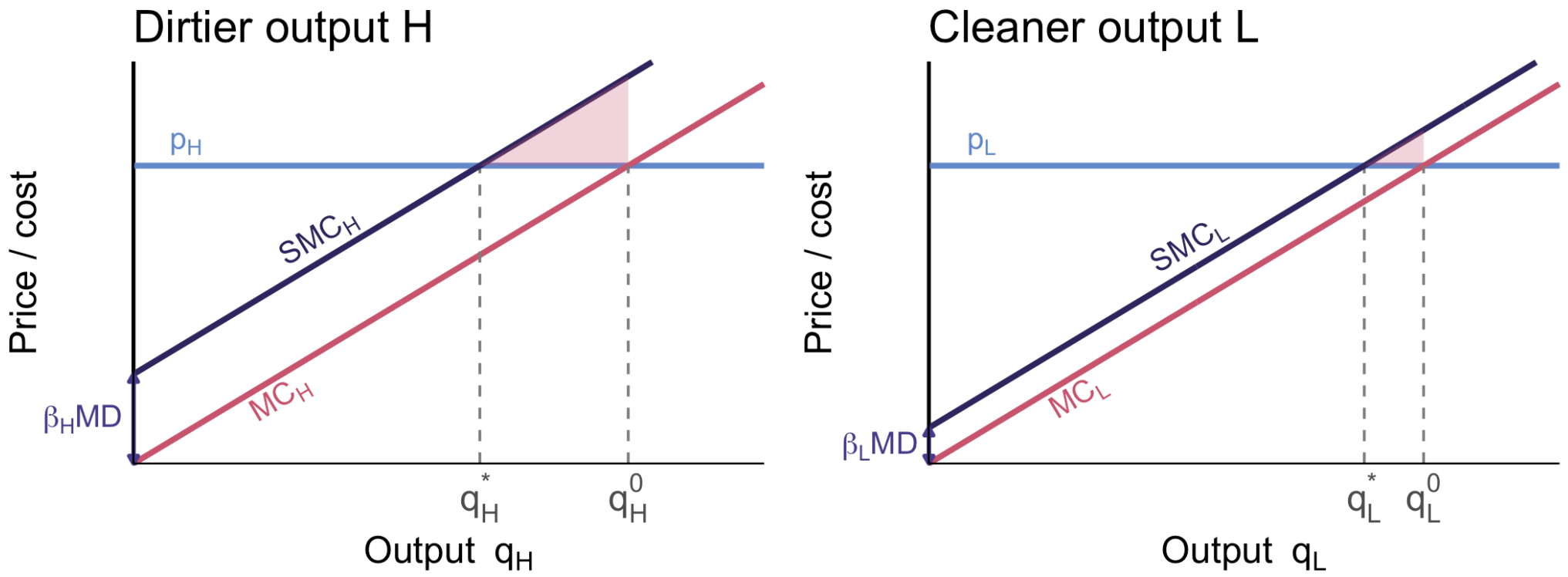
Intensity standards: baseline DWL

In H, the red triangle is DWL from output above q_H^*



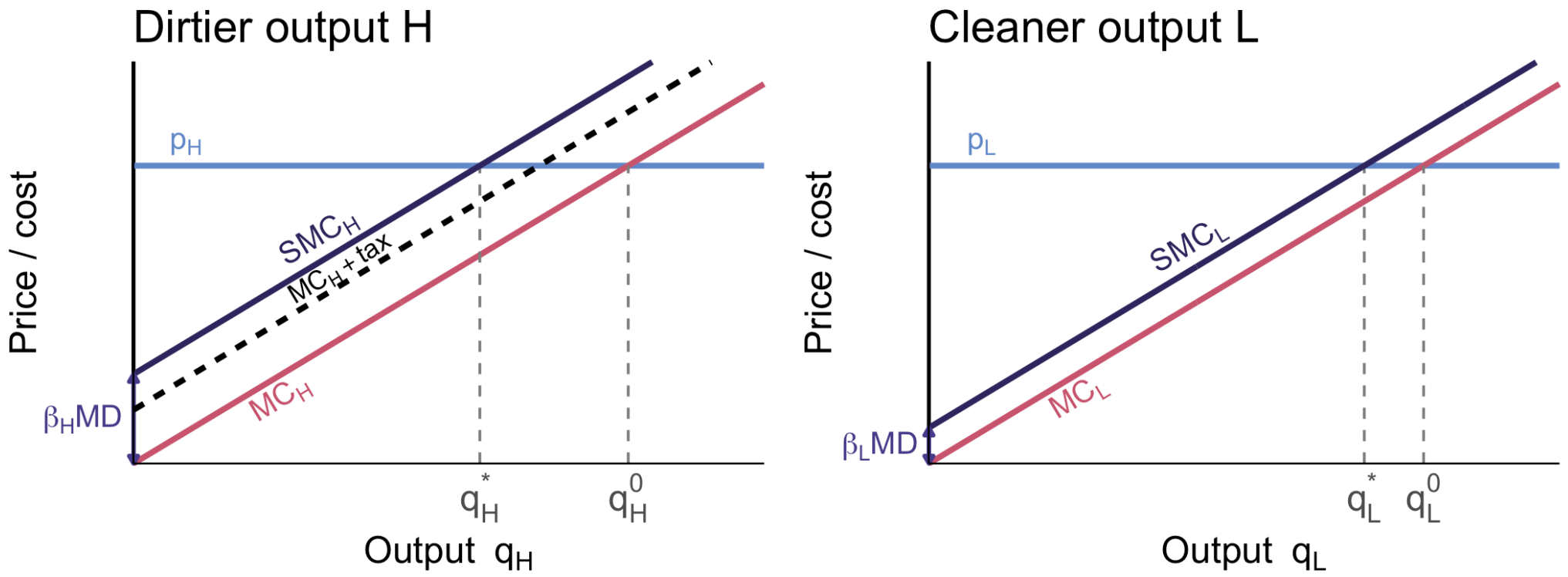
Intensity standards: baseline DWL

The cleaner good also pollutes, so L has a baseline DWL too



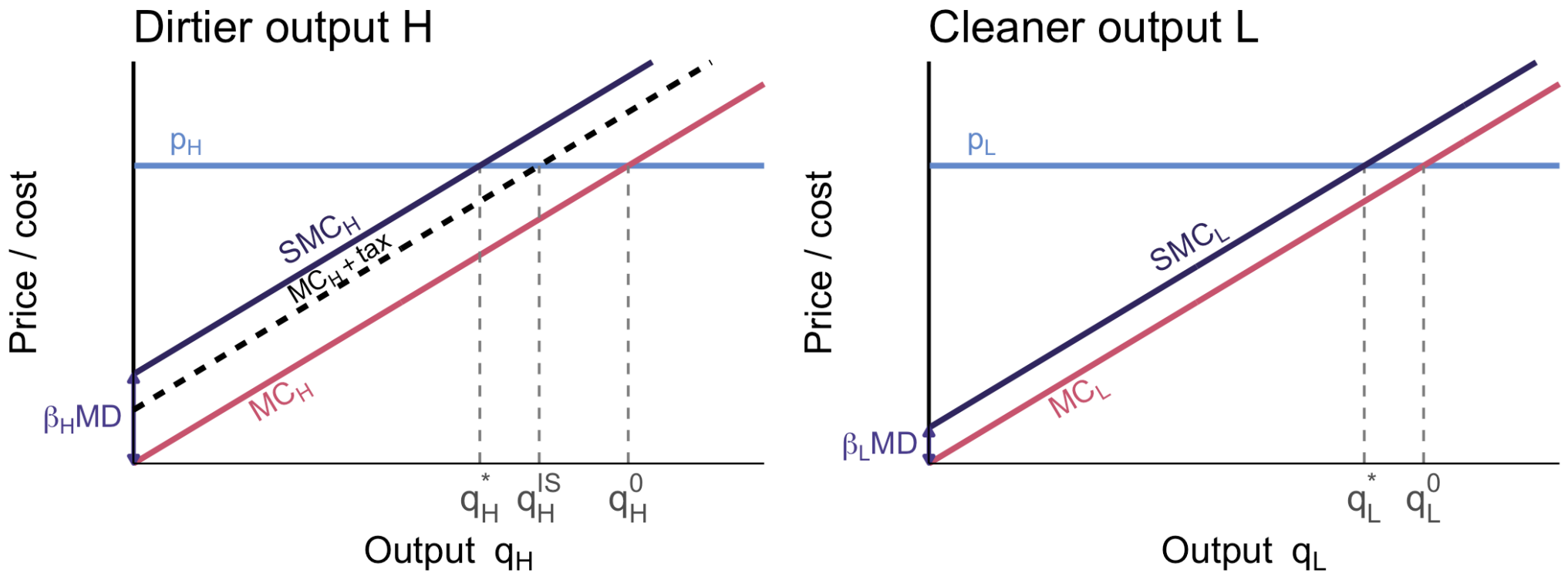
Intensity standards: the implicit tax

For H, the implicit tax shifts the firm's marginal cost up



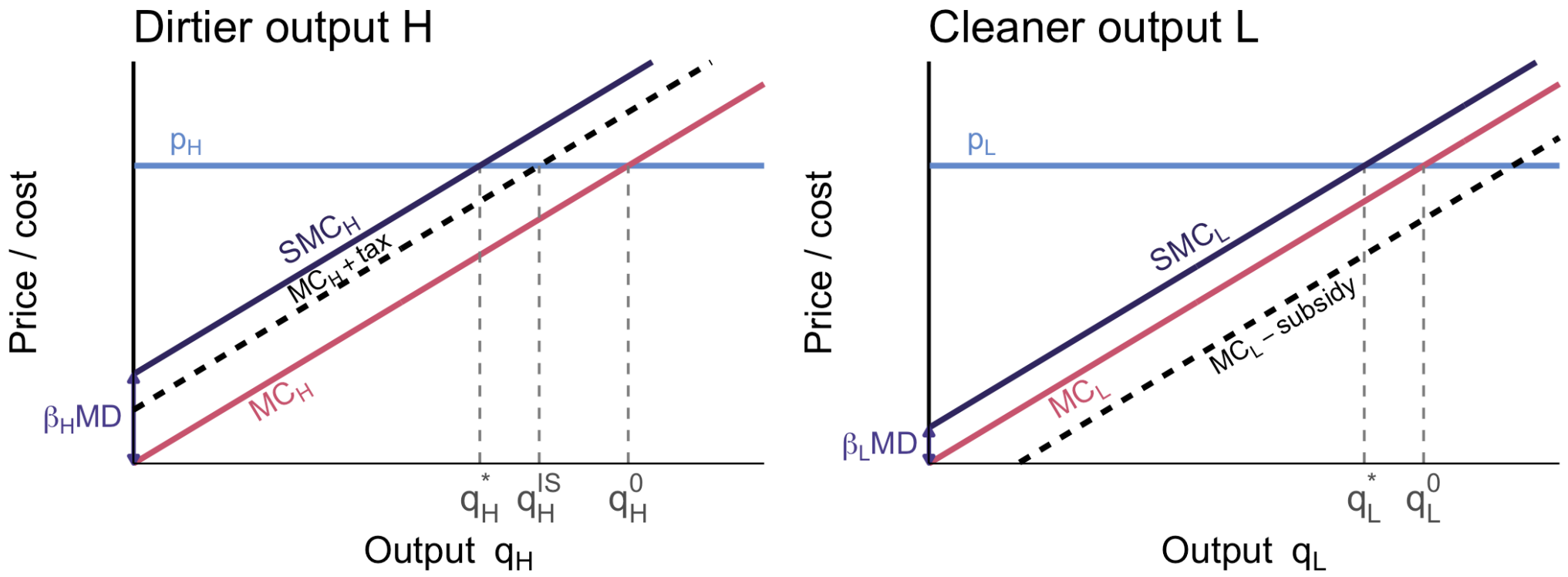
Intensity standards: the implicit tax

Output falls from q_H^0 to q_H^{IS}, toward q_H^*



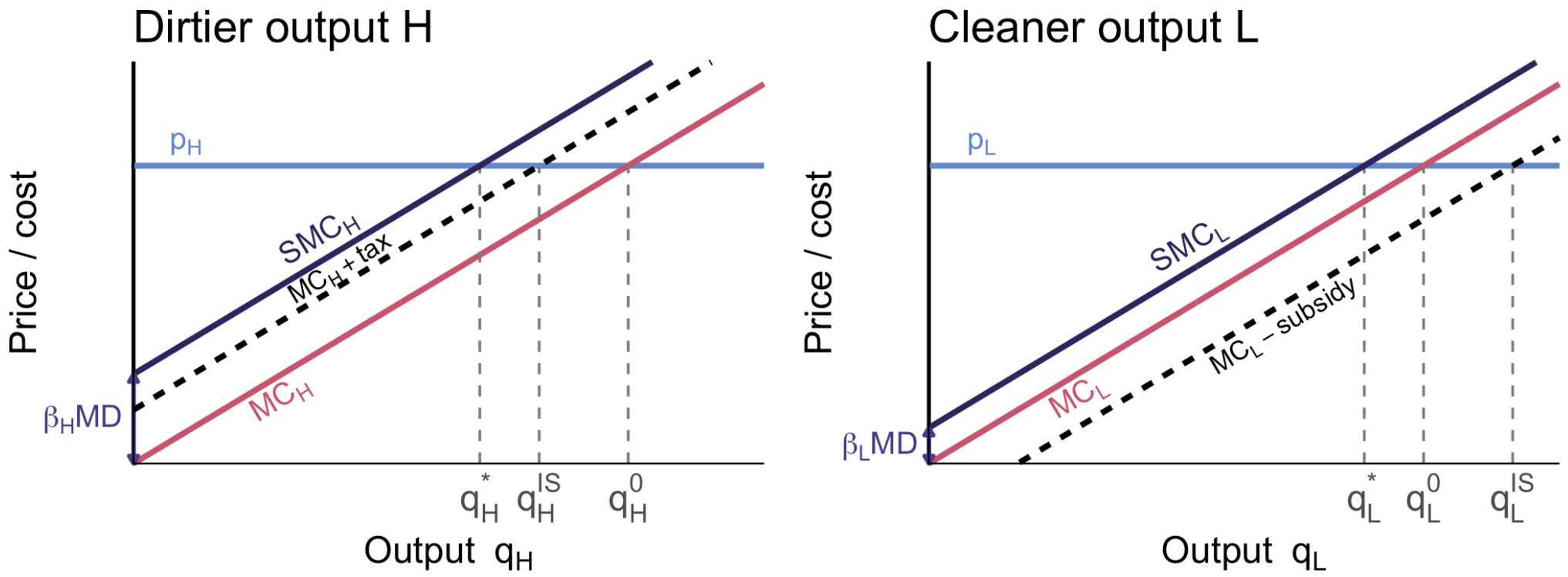
Intensity standards: the implicit subsidy

For L, the implicit subsidy shifts the firm's marginal cost down



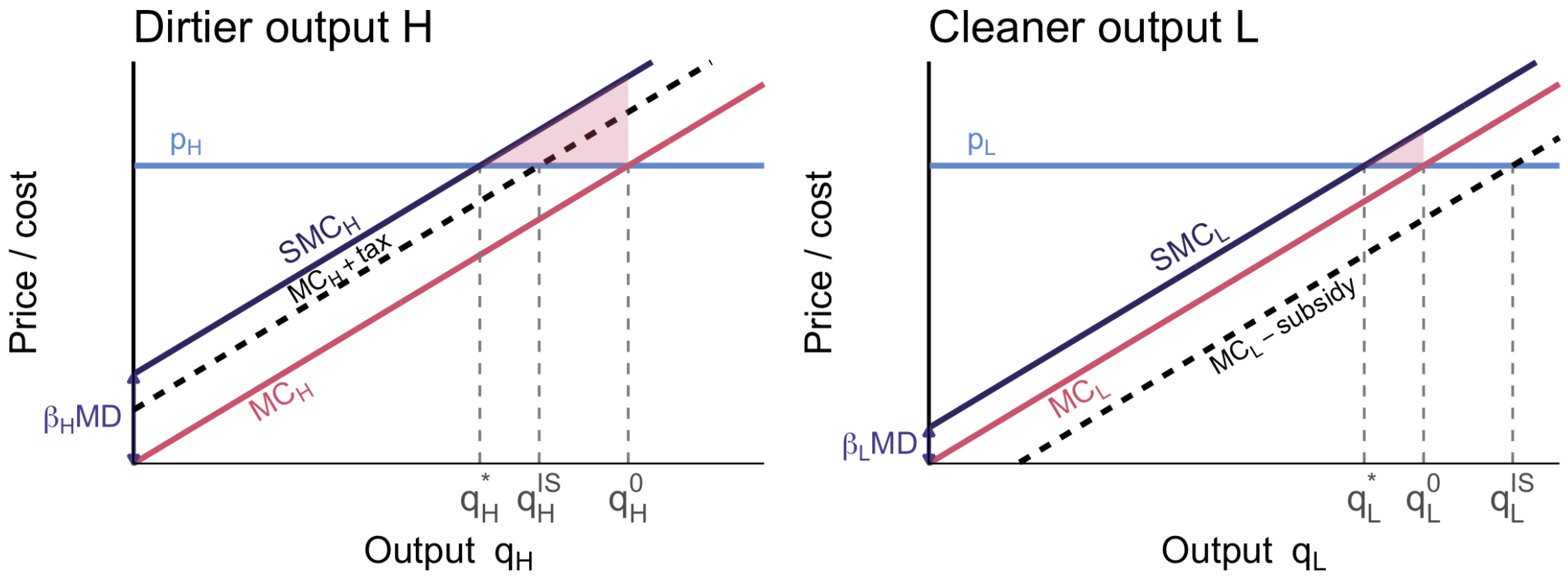
Intensity standards: the implicit subsidy

Output rises from q_L^0 to q_L^{IS}, farther from q_L^*



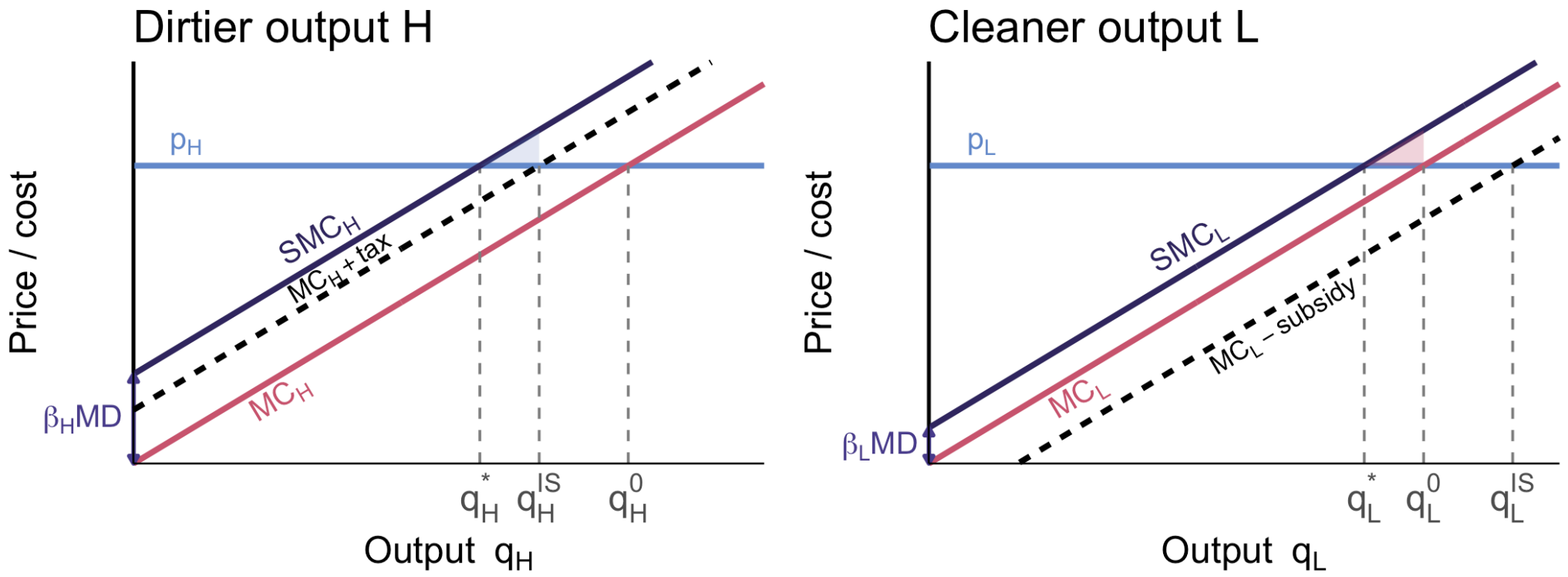
Intensity standards: baseline DWL

The red triangles show the DWL before the standard



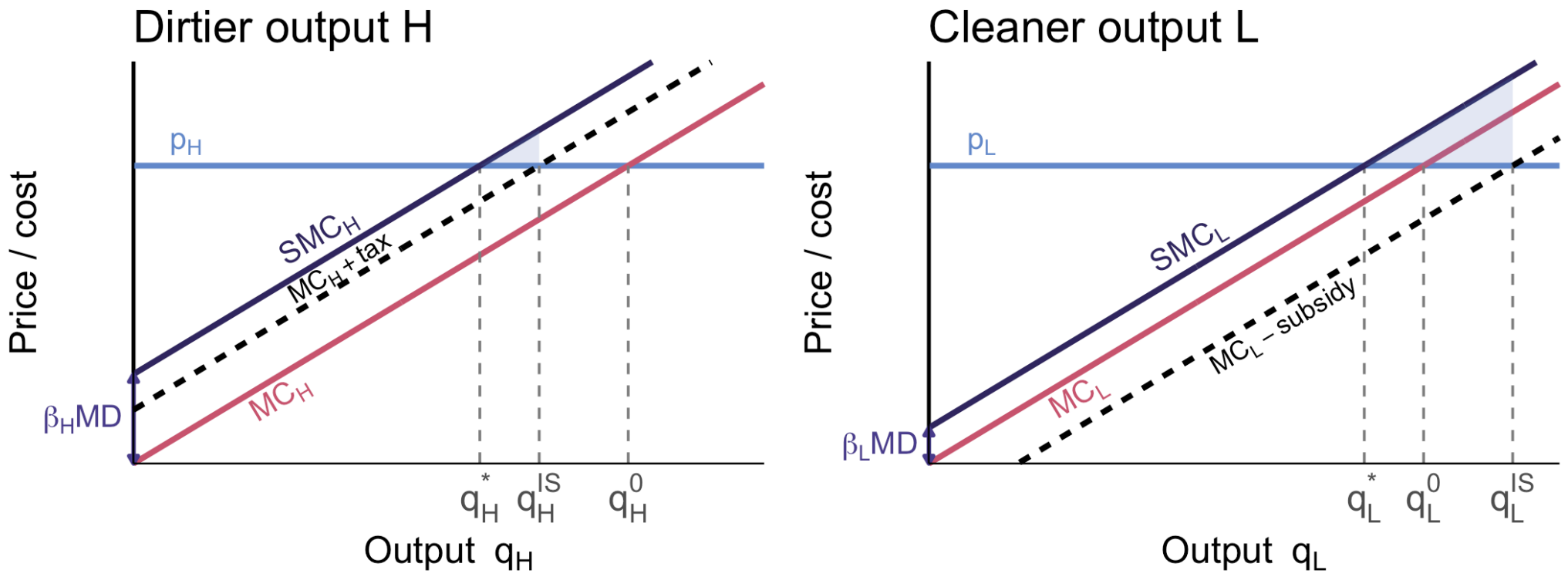
Intensity standards: DWL after the standard

In H, the smaller blue triangle shows the remaining DWL after the implicit tax



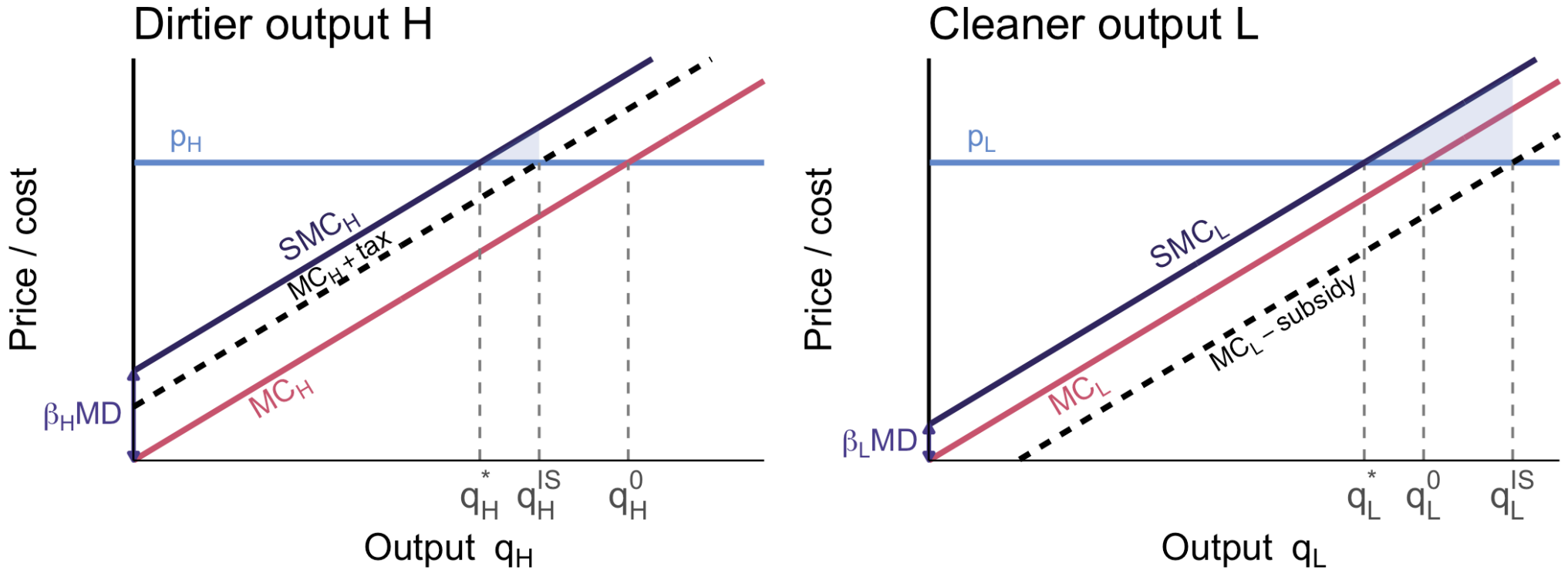
Intensity standards: DWL after the standard

In L, the larger blue triangle shows DWL rising with the implicit subsidy



Intensity standards: why DWL changes

Since DWL increases for L, we can't achieve the efficient outcome



Why this is second-best

An emissions tax prices all emissions at the same marginal damage

Why this is second-best

An emissions tax prices all emissions at the same marginal damage

An intensity standard instead mixes:

- an implicit tax on dirtier output
- an implicit subsidy on cleaner output

Why this is second-best

An emissions tax prices all emissions at the same marginal damage

An intensity standard instead mixes:

- an implicit tax on dirtier output
- an implicit subsidy on cleaner output

So it can improve on no policy, but usually cannot hit the socially optimal allocation

Intensity standards: scale vs composition

The policy helps on **composition**: firms shift away from dirtier output

Intensity standards: scale vs composition

The policy helps on **composition**: firms shift away from dirtier output

But it can hurt on **scale**: the implicit output subsidy encourages more total production

Intensity standards: scale vs composition

The policy helps on **composition**: firms shift away from dirtier output

But it can hurt on **scale**: the implicit output subsidy encourages more total production

Total emissions are:

$$E = \beta_H Q_H + \beta_L Q_L$$

Intensity standards: scale vs composition

The policy helps on **composition**: firms shift away from dirtier output

But it can hurt on **scale**: the implicit output subsidy encourages more total production

Total emissions are:

$$E = \beta_H Q_H + \beta_L Q_L$$

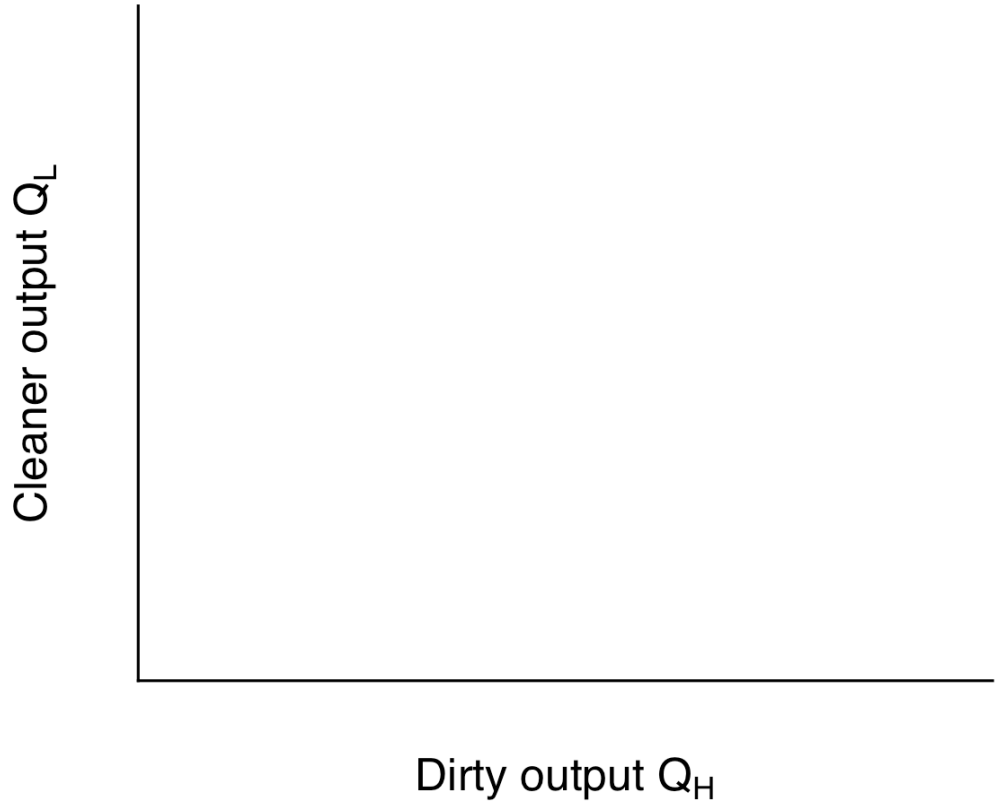
So total emissions can rise if the scale effect dominates the composition effect

Intensity standards: perverse incentives

Horizontal axis: dirty output Q_H

Vertical axis: cleaner output Q_L

Both outputs pollute: $0 < \beta_L < \beta_H$

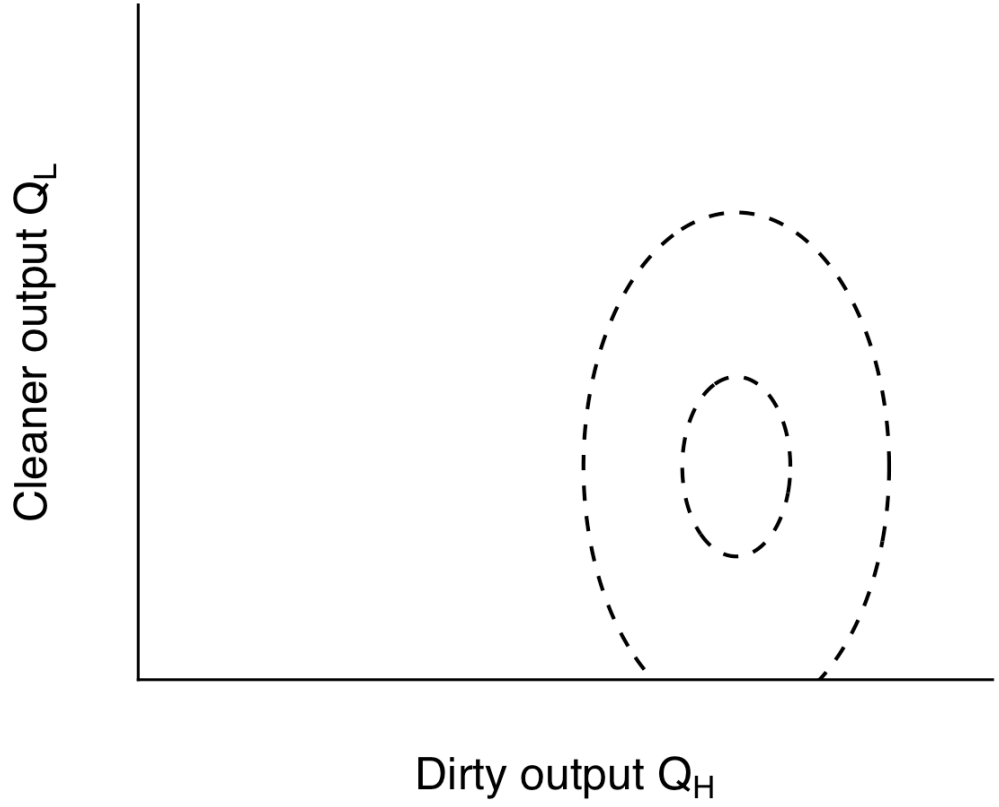


Intensity standards: perverse incentives

Iso-profit curves are the firm's indifference curves: equal profit along each oval

At given prices, rising MC eventually exceeds price, so producing more is not always better

This makes the curves ovals, with higher profit toward the center

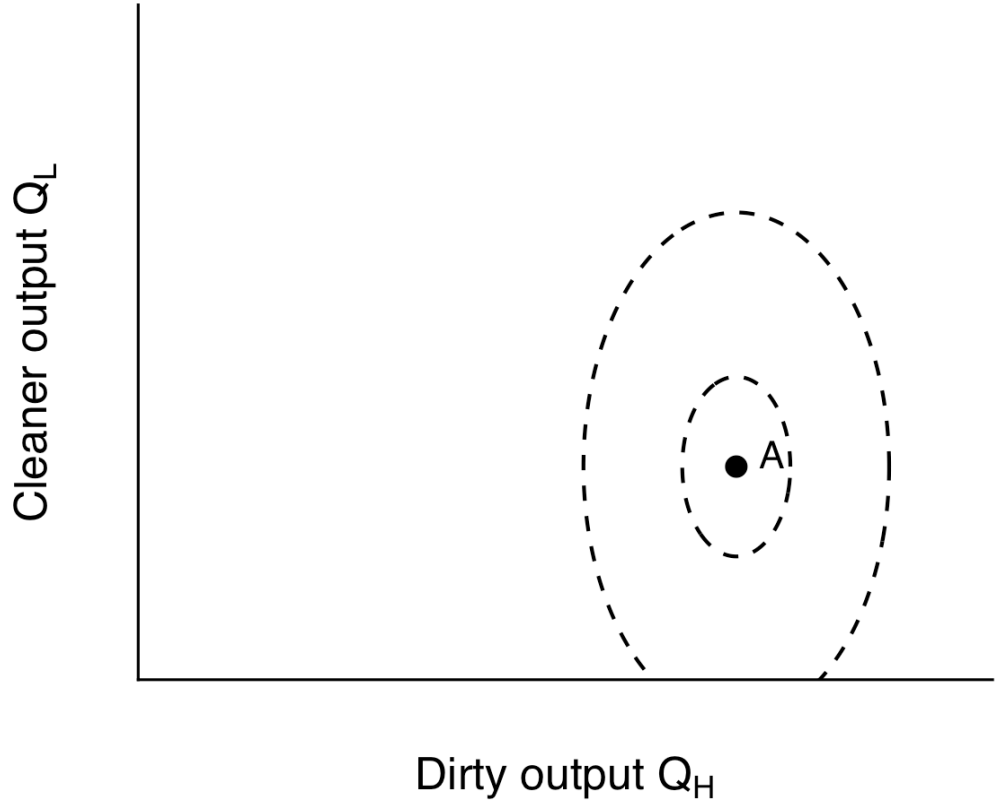


Intensity standards: perverse incentives

A is the **unregulated profit-maximizing choice**

It maximizes profit, ignoring emissions damages

Moving away from A lowers profit

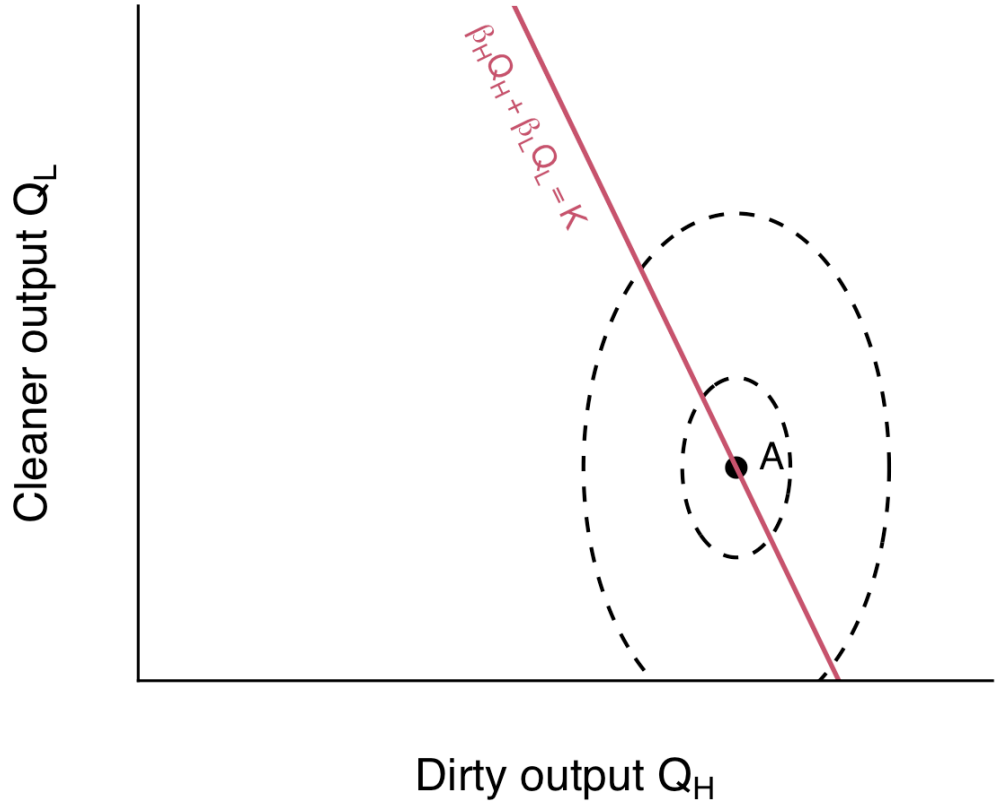


Intensity standards: perverse incentives

The red line holds **total emissions** fixed at A's level

$$\beta_H Q_H + \beta_L Q_L = K$$

Its slope is $-\beta_H / \beta_L$

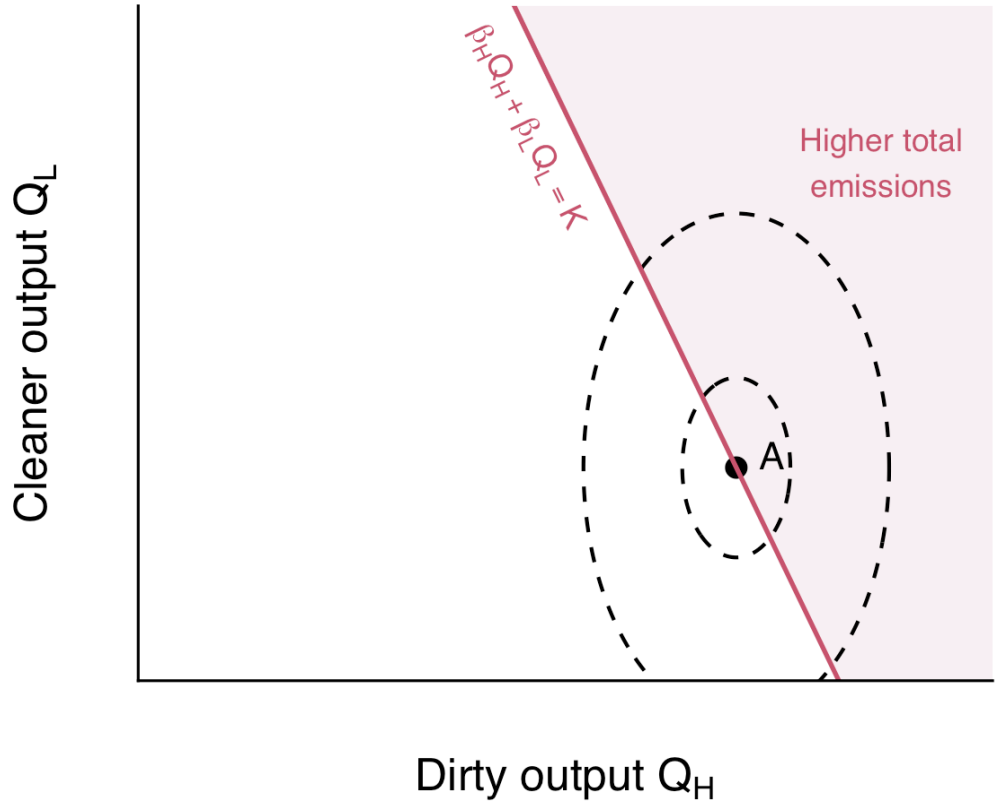


Intensity standards: perverse incentives

Above the red line: **more emissions than A**

Below the line: fewer emissions

The shaded region has $E > K$

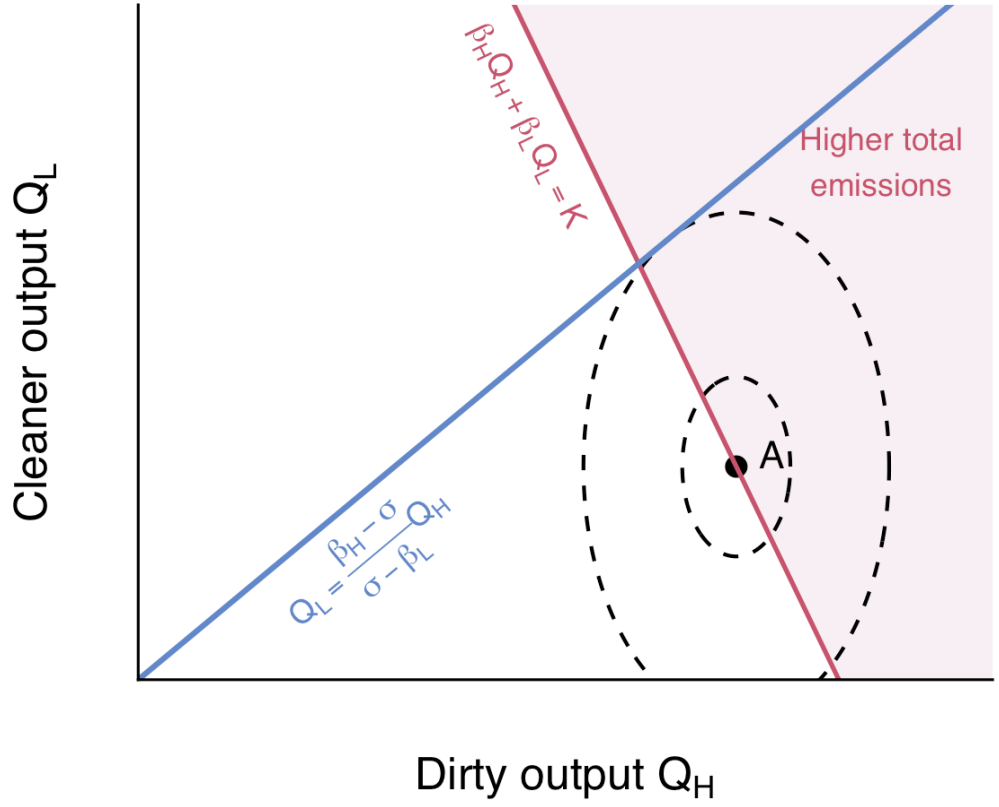


Intensity standards: perverse incentives

The blue line is the **binding standard**

$$Q_L = \frac{\beta_H - \sigma}{\sigma - \beta_L} Q_H$$

On the line, emissions intensity equals σ

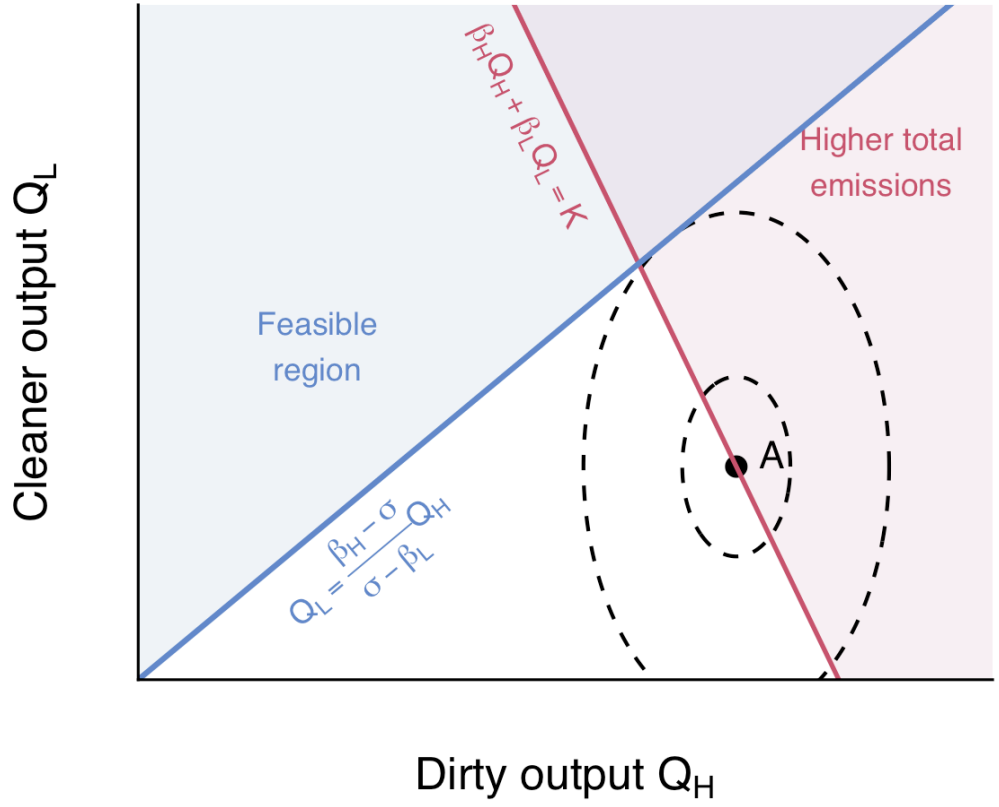


Intensity standards: perverse incentives

The blue line is the **binding standard**

$$Q_L = \frac{\beta_H - \sigma}{\sigma - \beta_L} Q_H$$

A violates the standard; feasible choices lie on or above the blue line

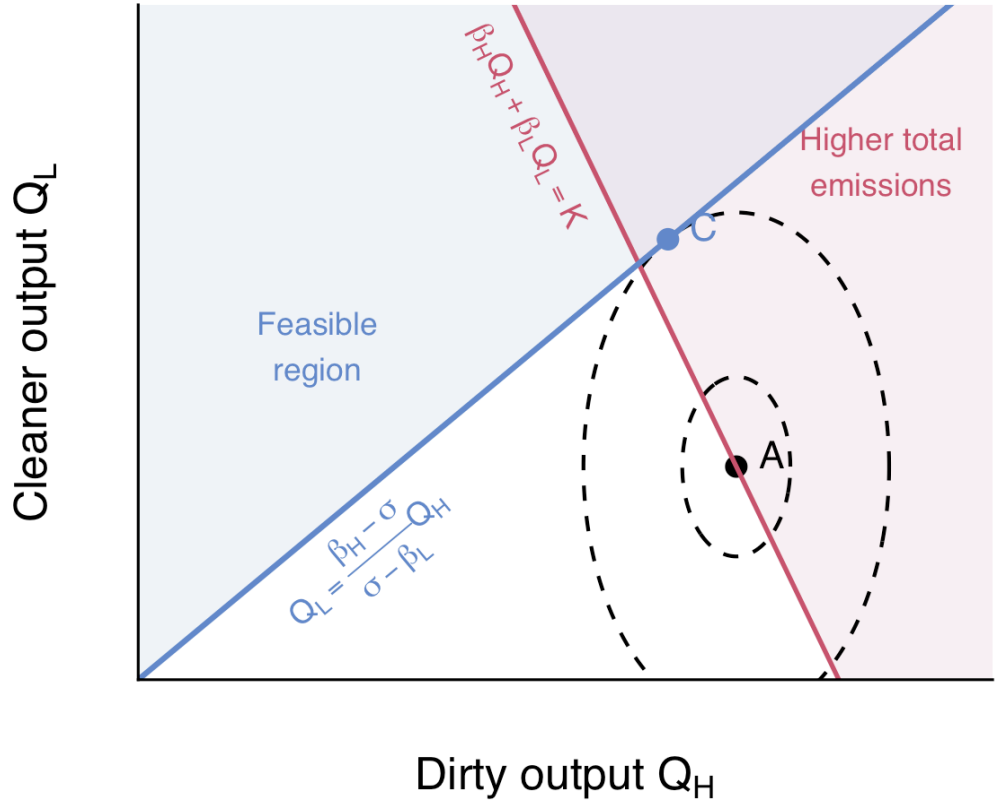


Intensity standards: perverse incentives

C is the **regulated profit-maximizing choice**

The highest attainable iso-profit curve is tangent to the standard at C

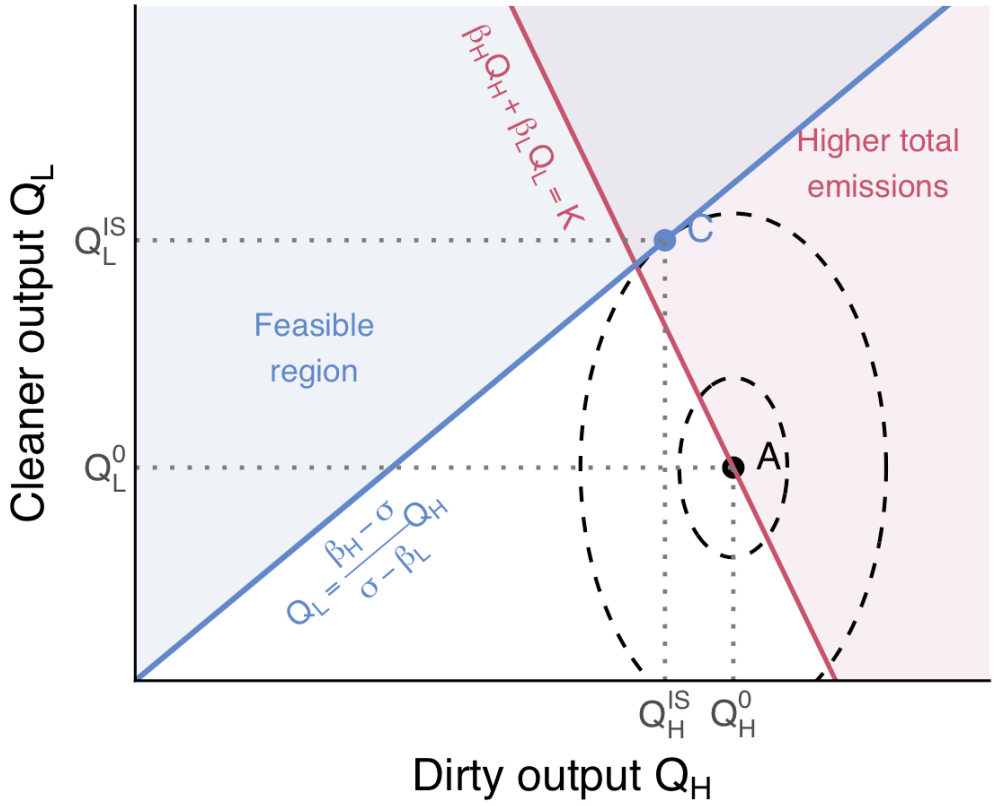
C lies above the red line: emissions increase



Intensity standards: intuition

Dirty output falls: $Q_H^{IS} < Q_H^0$

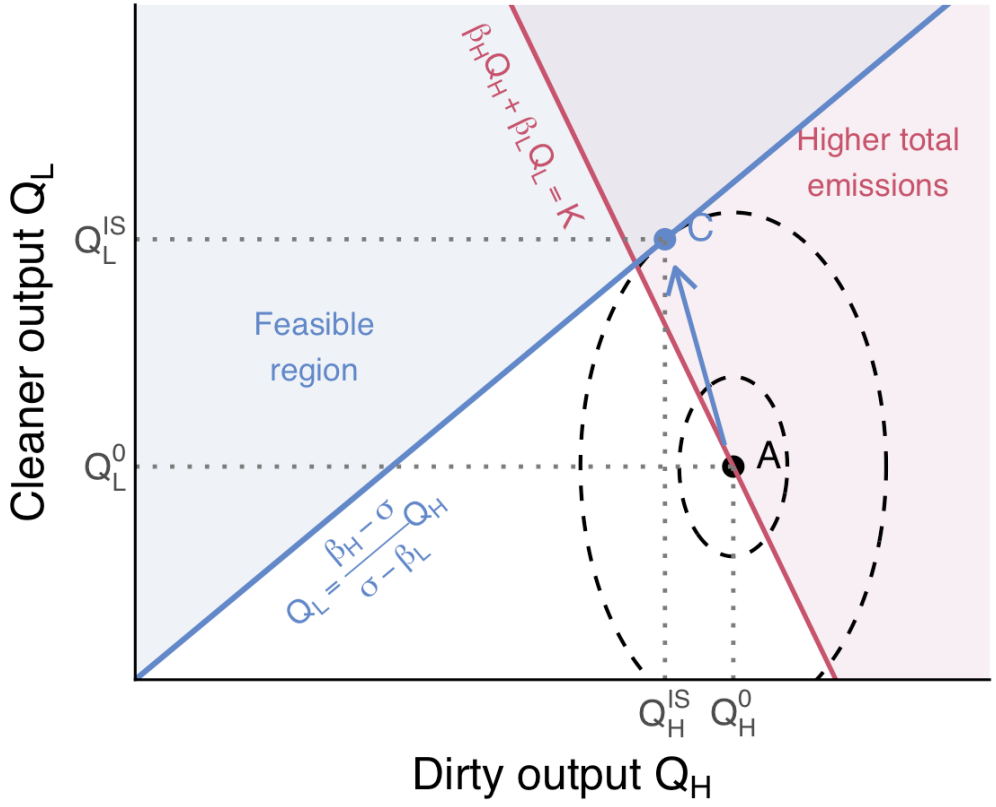
Cleaner output rises: $Q_L^{IS} > Q_L^0$



Intensity standards: intuition

Tall ovals mean expanding cleaner output reduces profit less than an equal cut in dirty output

Here, **total output rises** while emissions intensity falls

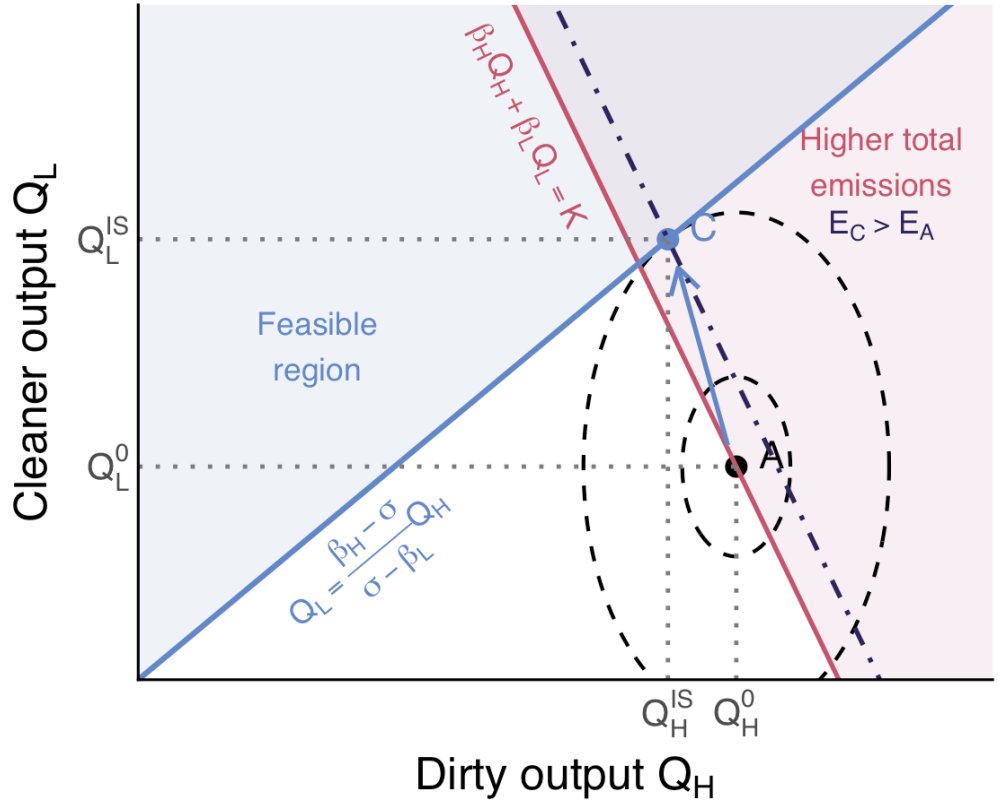


Intensity standards: intuition

At C, the purple emissions line lies above A's red line

Extra emissions from L exceed the savings from H

Total emissions rise even though intensity falls

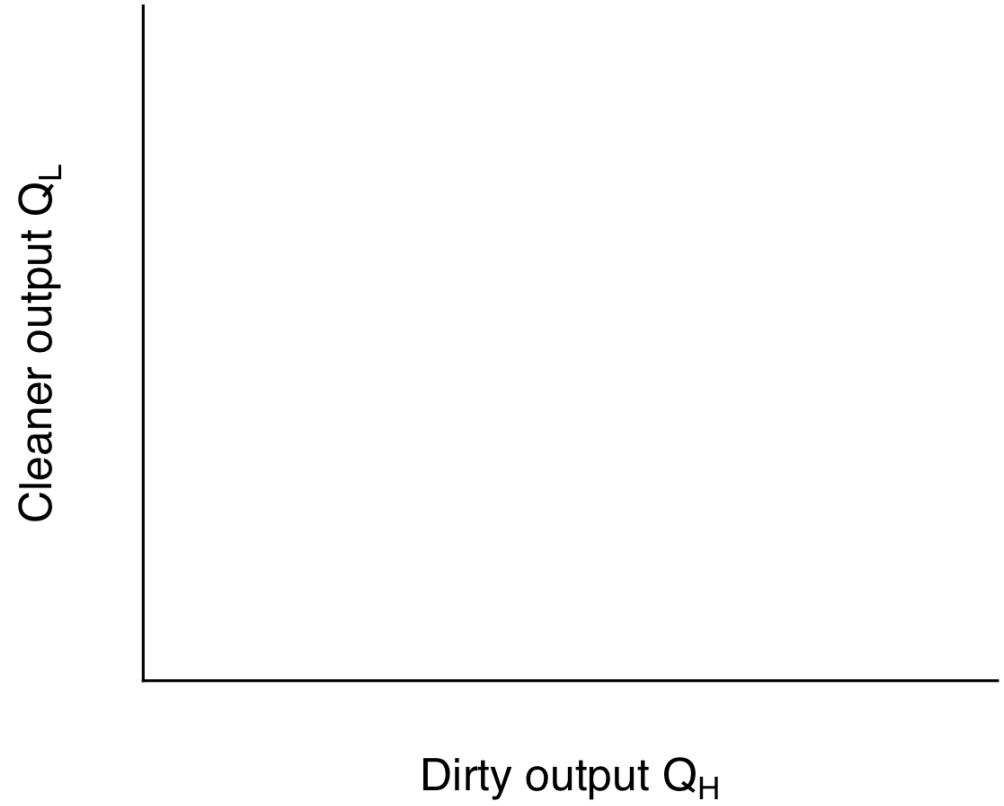


Intensity standards: emissions can fall

Horizontal axis: dirty output Q_H

Vertical axis: cleaner output Q_L

Keep the same emissions rates and standard

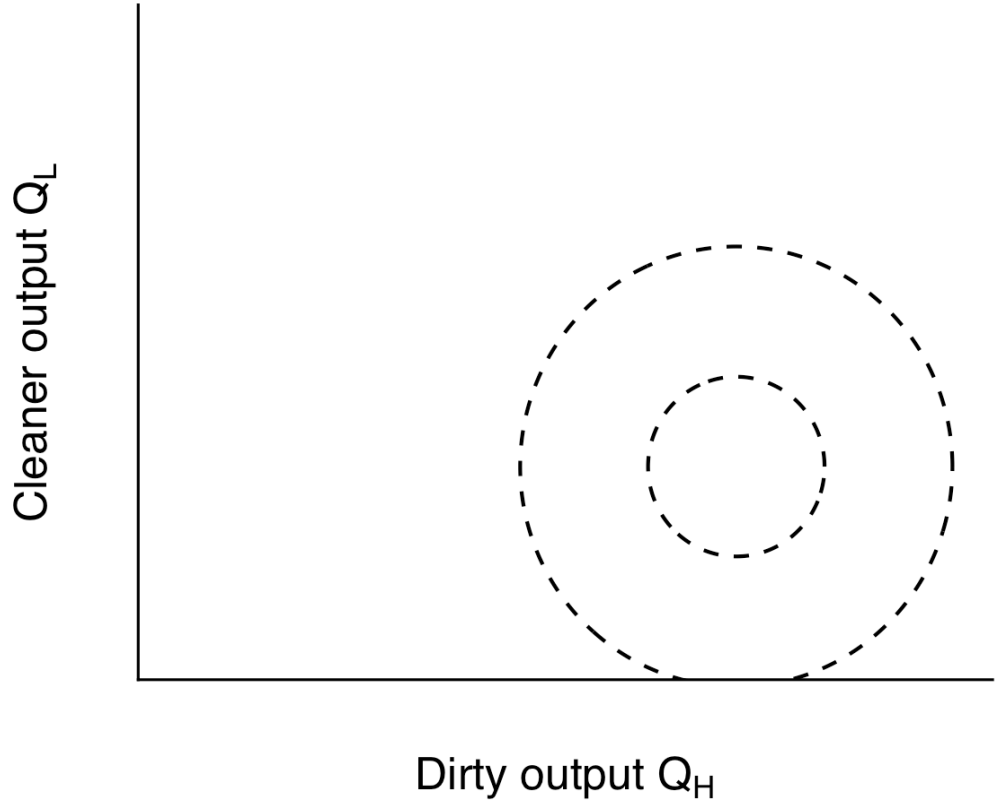


Intensity standards: emissions can fall

These fatter **iso-profit curves** are the firm's indifference curves: equal profit along each oval

At given prices, rising MC eventually exceeds price, so producing more is not always better

This makes the curves ovals, with higher profit toward the center

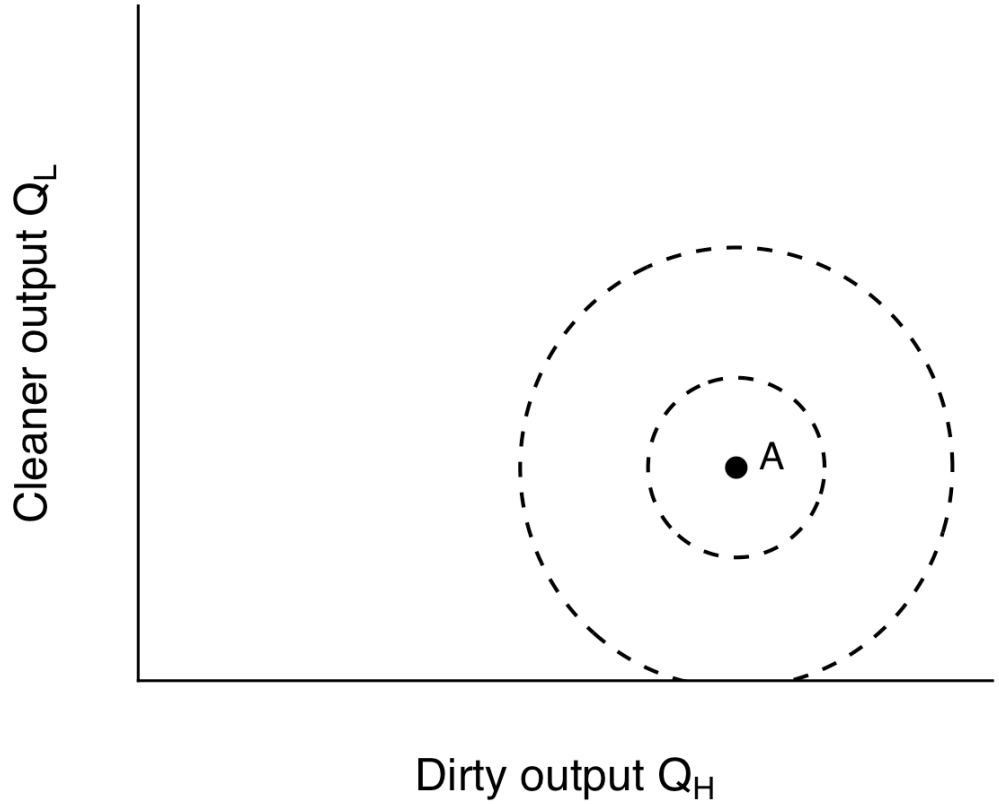


Intensity standards: emissions can fall

A is the **unregulated profit-maximizing choice**

It maximizes profit, ignoring emissions damages

Moving away from A lowers profit

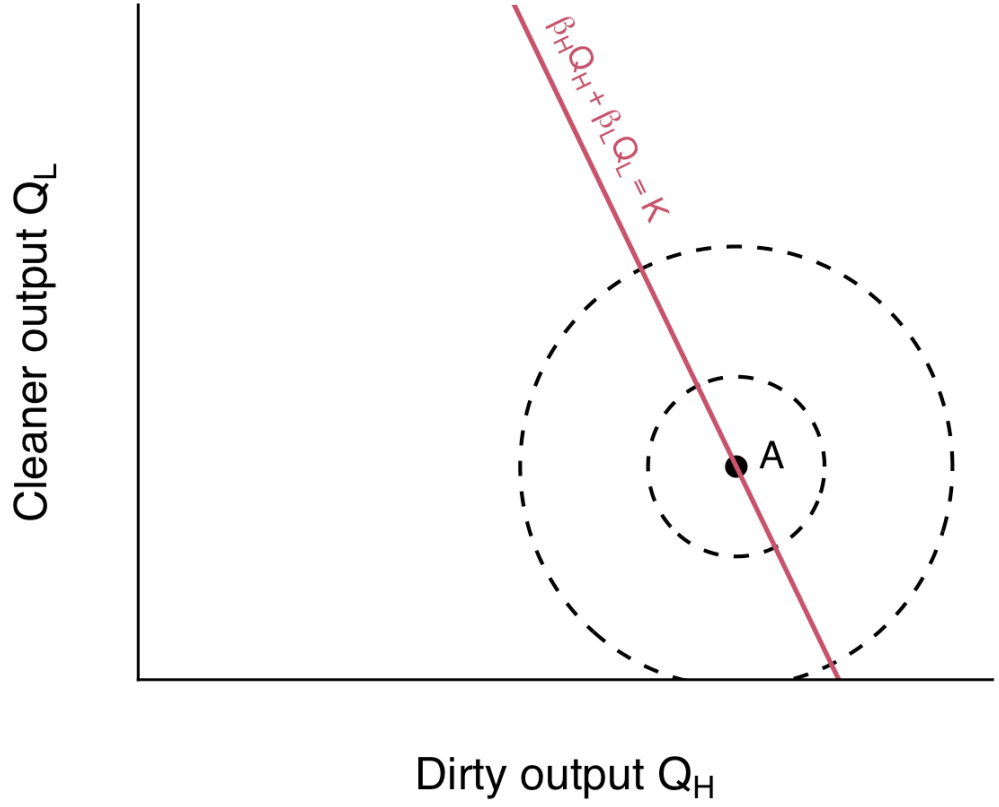


Intensity standards: emissions can fall

The red line holds **total emissions** fixed at A's level

$$\beta_H Q_H + \beta_L Q_L = K$$

Its slope is $-\beta_H / \beta_L$

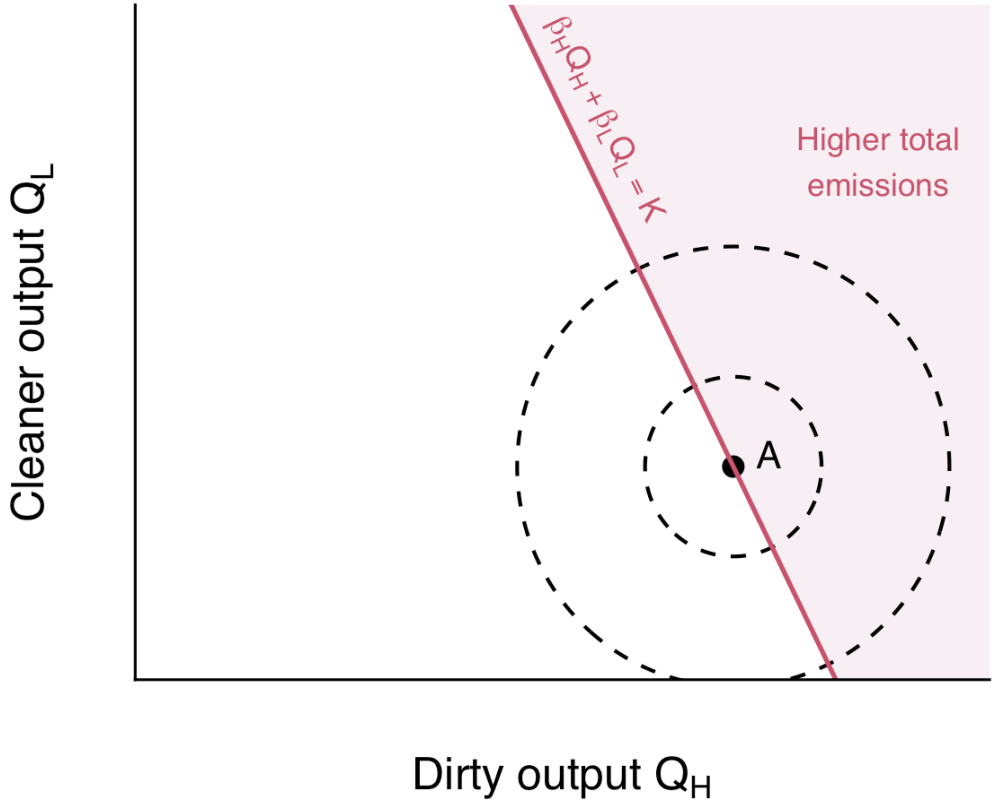


Intensity standards: emissions can fall

Above the red line: **more emissions than A**

Below the line: fewer emissions

The shaded region has $E > K$

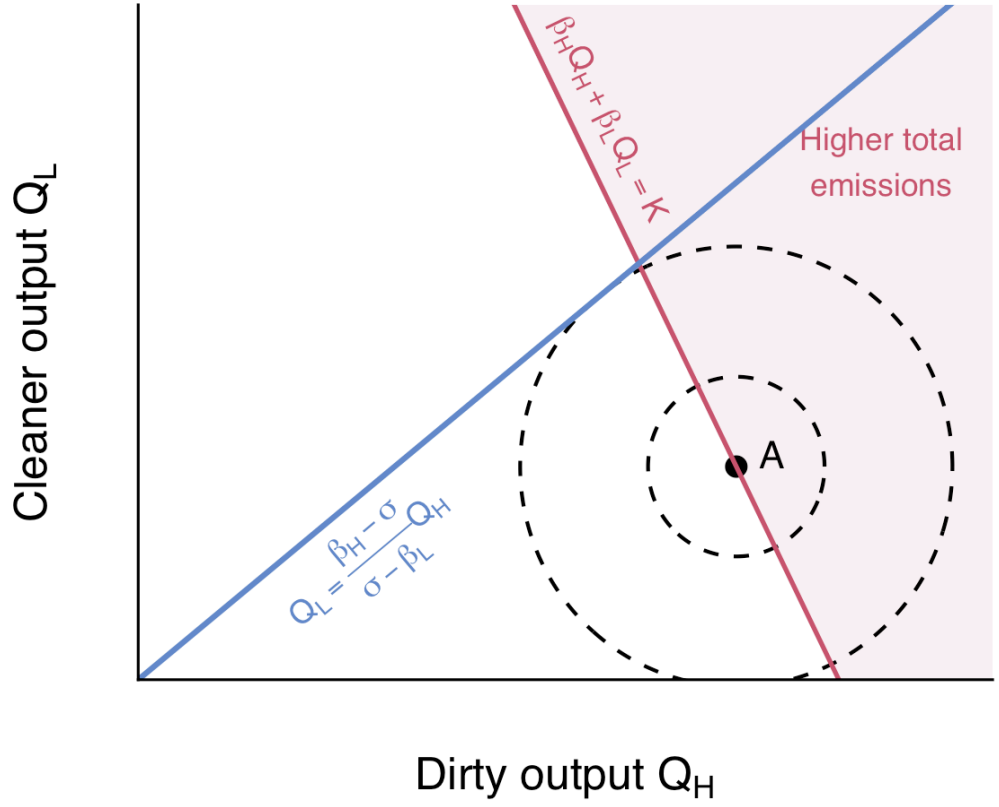


Intensity standards: emissions can fall

The blue line is the **binding standard**

$$Q_L = \frac{\beta_H - \sigma}{\sigma - \beta_L} Q_H$$

On the line, emissions intensity equals σ

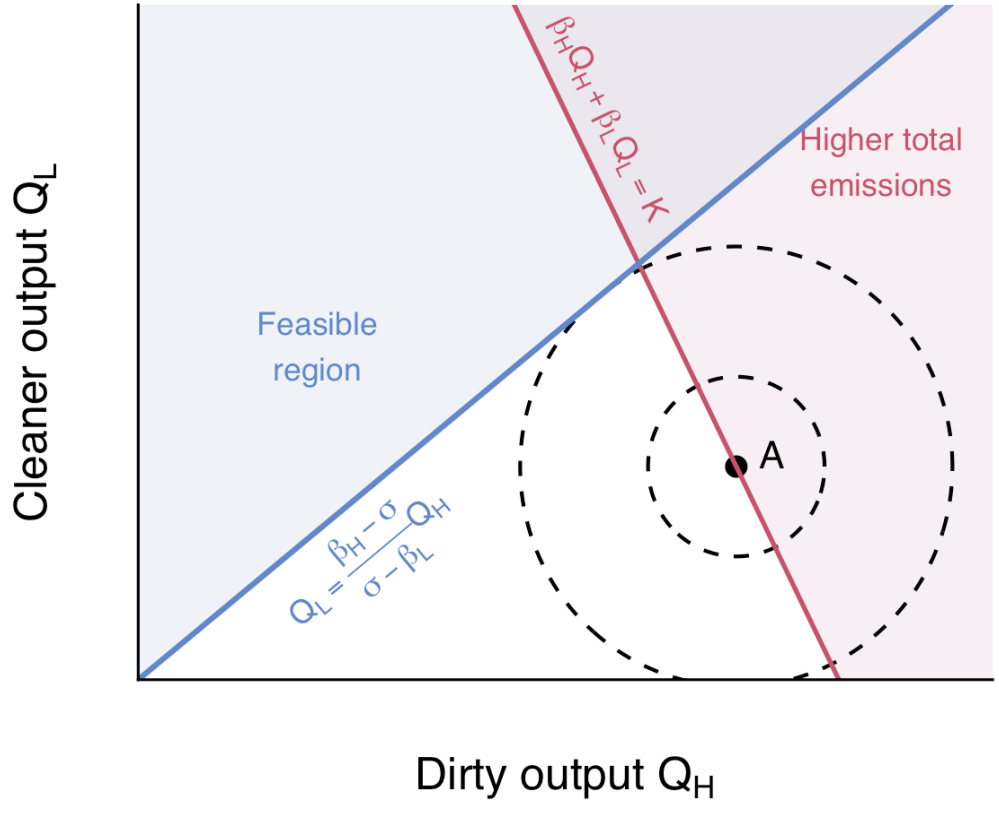


Intensity standards: emissions can fall

The blue line is the **binding standard**

$$Q_L = \frac{\beta_H - \sigma}{\sigma - \beta_L} Q_H$$

A violates the standard; feasible choices lie on or above the blue line

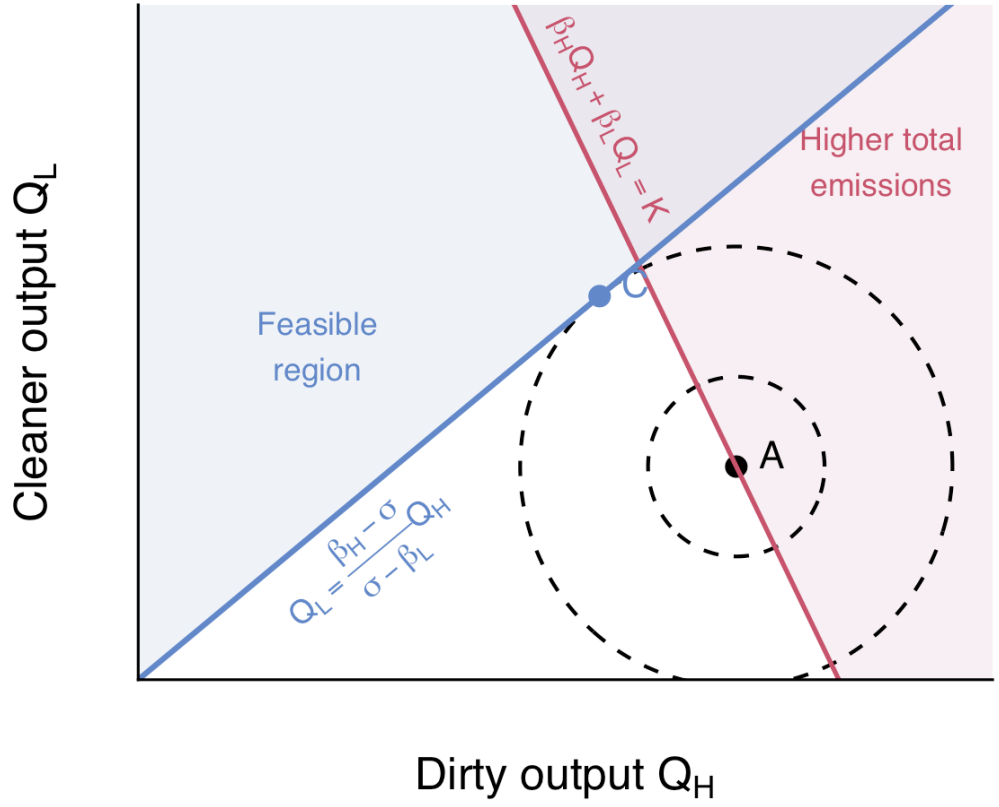


Intensity standards: emissions can fall

C is the **regulated profit-maximizing choice**

The highest attainable iso-profit curve is tangent to the standard at C

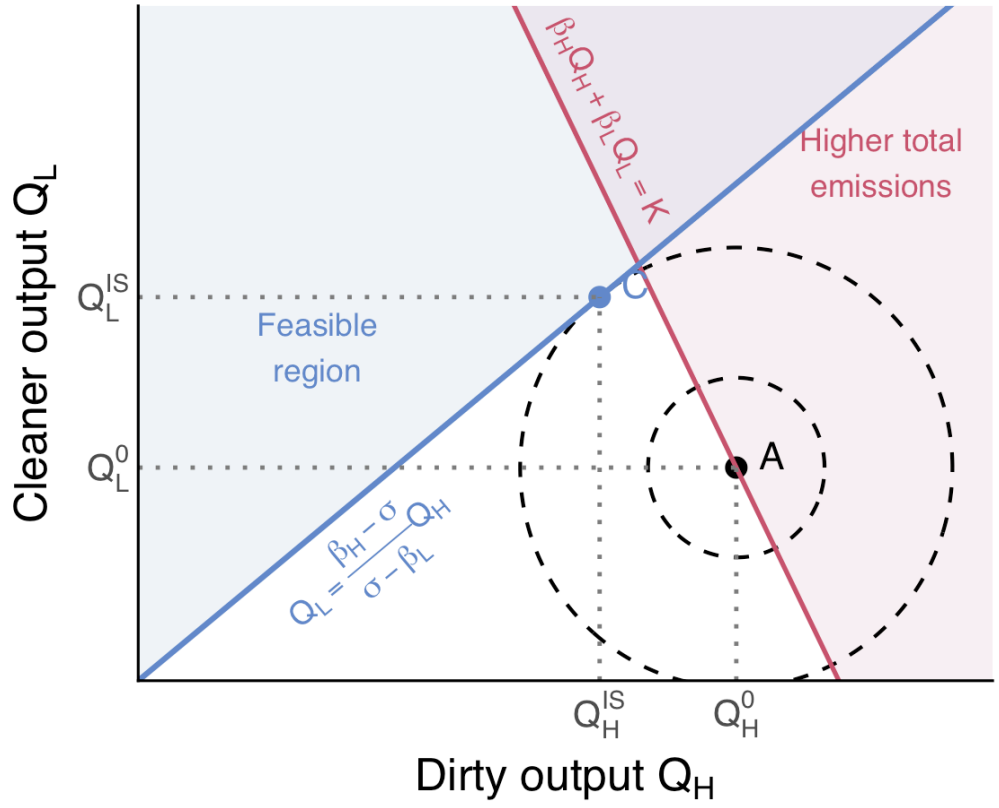
C lies below the red line: emissions decrease



Intensity standards: emissions fall

Dirty output falls: $Q_H^{IS} < Q_H^0$

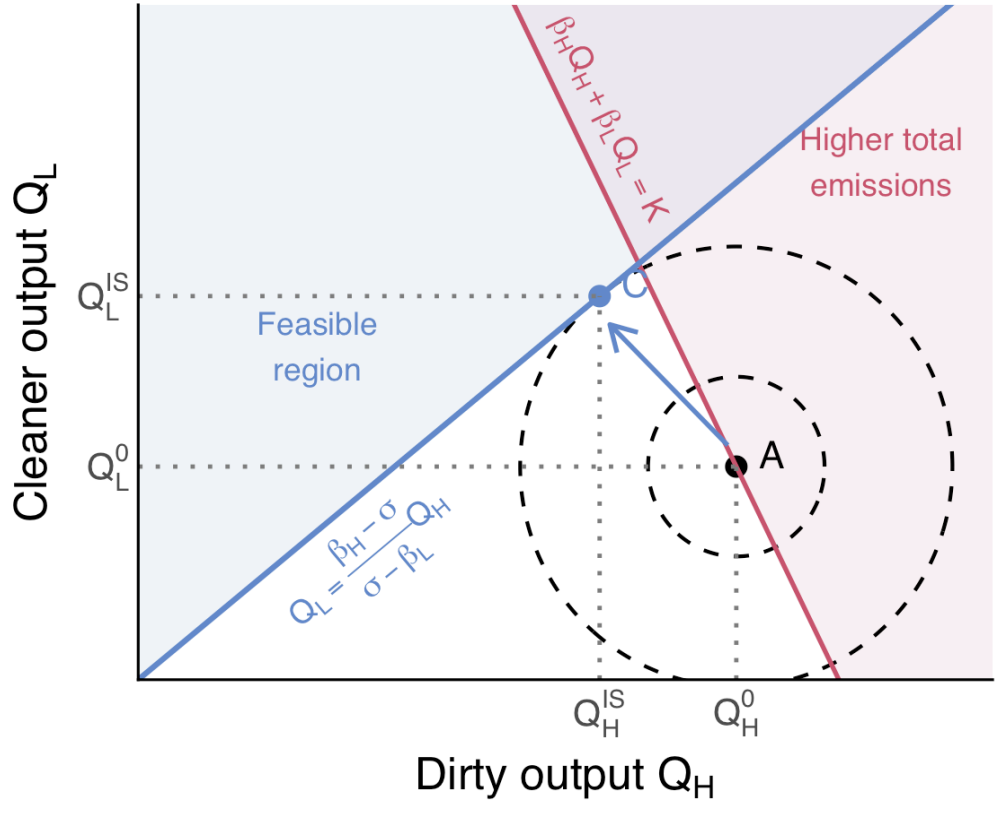
Cleaner output rises: $Q_L^{IS} > Q_L^0$



Intensity standards: emissions fall

Fatter ovals make cutting dirty output less costly relative to expanding cleaner output

Here, the **shift toward cleaner output** dominates the increase in total output



Intensity standards: emissions fall

At C, the purple emissions line lies below A's red line

Emissions saved from H exceed the extra emissions from L

Total emissions fall along with emissions intensity

